

Designing an Inclusive Digital Marketplace for Livestock: Facilitating Bargaining and Accessibility in Southern Telangana

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by

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**PDPM Indian Institute of Information Technology,
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DEDICATION

To my mother, father and brother.

CERTIFICATE

This is to certify that the thesis entitled, “**Designing an Inclusive Digital Marketplace for Livestock: Facilitating Bargaining and Accessibility in Southern Telangana**”, submitted by **Mr. Pavan** , Roll No. **2123**, in partial fulfilment of the requirements for the award of **Master of Design (M.Des.)** degree at PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur, is an authentic record of work carried out by him under my supervision and guidance.

To the best of my knowledge, the matter embodied in this thesis has not been submitted elsewhere to any other university or institute for the award of any other degree.

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Finally, I thank my family and friends for their constant encouragement and support during the course of this thesis and my M.Des journey.

Pavan

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ABSTRACT

Livestock trading in Southern Telangana continues to be conducted largely through weekly physical markets known as Santhas, where buyers and sellers rely on face-to-face bargaining rather than fixed pricing. This informal system, while socially embedded and trusted, imposes considerable costs in travel, time, transparency and market reach, particularly for low-literacy, Telugu-speaking rural traders. Existing digital livestock and agricultural marketplaces, including e-NAM and comparable commercial applications, have tended to prioritise fixed-price listings and text-heavy interfaces, offering little support for negotiation, vernacular interaction, or trust-building mechanisms suited to this user group.

This thesis followed a human-centred design process to understand these practices and to design and evaluate an inclusive digital marketplace, Sukraven, that preserves bargaining while improving accessibility, transparency and trust. A mixed-methods field study was conducted in December 2025 with 82 participants (47 buyers, 35 sellers) across the Nagarkurnool, Wanaparthy and Mahabubnagar districts, combining direct market observation with a structured, orally-administered questionnaire covering participant profile, market experience and technology readiness. Findings showed that 76.8 per cent of participants considered bargaining very important to obtaining a fair price, that lack of transportation, price transparency and broker influence were the most frequently cited problems, and that while 69.5 per cent of participants owned a smartphone, only 24.4 per cent had ever used any online platform to trade livestock.

These insights informed the design of Sukraven, a Telugu- and English-language mobile application supporting verified profiles, four-angle livestock photographs and video, digital bargaining through offer and counter-offer, a Bayana (token-deposit) reservation mechanism, and delivery coordination. The resulting high-fidelity prototype was evaluated with 18 independent participants (10 buyers, 8 sellers) in May 2026 using the System Usability Scale (SUS), a Trust Perception Scale and a User Satisfaction Scale. The prototype achieved a mean SUS score of 80.42 out of 100, corresponding to a 'Good' usability rating, a mean trust-perception score of 4.31 out of 5, and a mean user-satisfaction score of 4.29 out of 5.

The thesis concludes that digitising the negotiation-centred structure of the Santha, rather than replacing it with fixed pricing, together with vernacular language support and visual, low-literacy-friendly interface patterns, can produce a livestock marketplace that rural buyers and sellers find both usable and trustworthy. The study also identifies limitations, including the

small usability-evaluation sample, an unresolved item of reverse-worded response validity in the SUS data (Section 6.2), and institutional ethics review that remains pending at the time of submission following retrospective referral to the PDPM IIITDM Jabalpur Institutional Ethics Committee, and outlines directions for future work.

Keywords: human-centred design, inclusive design, digital marketplace, livestock trading, bargaining, System Usability Scale, rural India, Telangana

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Nomenclature / Abbreviations

- HCD — Human-Centred Design
- IEC — Institutional Ethics Committee
- M.Des — Master of Design
- SUS — System Usability Scale
- UPI — Unified Payments Interface
- e-NAM — Electronic National Agricultural Market
- Santha — a weekly, physical livestock market held in Southern Telangana
- Bayana — a partial (token) advance payment that reserves a livestock listing for the buyer
- Sukraven — the mobile-application prototype designed and evaluated in this thesis

Chapter 1 Introduction

1.1 Background

Livestock trading forms a significant part of the rural economy across Southern Telangana, where cattle, buffalo, sheep and goats are bought and sold primarily through weekly livestock markets known locally as Santhas. Districts such as Nagarkurnool, Wanaparthy and Mahabubnagar host several such markets, including the Kaverammampeta, Pebbair, Bijinapally and Kalwakurthy Santhas, which draw buyers and sellers from surrounding villages on fixed days of the week. Trading in these markets follows a largely informal, face-to-face structure: animals are inspected physically, prices are arrived at through open bargaining rather than fixed listings, and payment is typically made in cash at the point of sale.

This informal structure has persisted because it accommodates the practical needs of the trade: buyers can physically inspect an animal's health and temperament before committing to a price, sellers retain the ability to negotiate rather than accept a posted figure, and transactions are grounded in face-to-face trust built over repeated market visits. At the same time, the system carries real costs. Buyers and sellers must travel, sometimes considerable distances, on a fixed market day regardless of whether a suitable animal or buyer is available. Price information is not transparent across markets, which leaves room for brokers and intermediaries to influence outcomes. Market-goers with lower literacy or limited exposure to formal price-setting mechanisms are particularly exposed to unfavourable terms.

Digital marketplaces have transformed agricultural and livestock trade in several respects elsewhere in India, most visibly through the Government of India's electronic National Agricultural Market (e-NAM), which seeks to unify agricultural commodity markets under a common online trading platform (Manogna, 2025). However, awareness and usage of such platforms among smallholder farmers and traders remain uneven; even where digital literacy is reasonably high, awareness of specific platforms, comfort with the underlying transaction process and perceived risk continue to limit adoption (Manogna, 2025), a pattern corroborated by district-level studies of e-NAM specifically, which report registration difficulty, weak digital literacy and continued trust in traditional intermediaries as recurring barriers to adoption (Gautam, Srivastava & Bohra, 2022), and by broader survey evidence that education level and peer adoption within a village are stronger predictors of farmers' access to digital agricultural and financial services than infrastructure availability alone (Sharma, Katoch, Mohapatra, Kaur

& Kumar, 2025). Livestock-specific digital marketplaces in India, meanwhile, have tended to replicate a fixed-price, catalogue-style e-commerce structure that is poorly suited to a trading culture in which bargaining is central and where many users are low-literacy, vernacular-language speakers who are unfamiliar with smartphone interaction patterns designed primarily for urban, English-literate audiences (Medhi et al., 2011).

This thesis addresses that gap. It reports on a human-centred design research project undertaken to understand livestock-trading practices, bargaining behaviour and technology readiness among rural buyers and sellers in Southern Telangana, and to design and evaluate a mobile application, named Sukraven, that digitises the Santha experience while preserving its negotiation-centred character.



Figure 1.1: Livestock market (Pashuvula Santha) scene in Kalwakurthy Santha .



Figure 1.2: Livestock market (Pashuvula Santha) scene in Pebbair Santha.

1.2 Problem Statement

Existing digital marketplaces for livestock and agricultural produce in India are generally designed around fixed pricing, text-dense interfaces and formal registration flows. These design choices are at odds with the lived experience of rural livestock traders in Southern Telangana, for whom bargaining is integral to arriving at a fair price, and for whom low formal education, limited English literacy and inconsistent smartphone familiarity make conventional e-commerce interaction patterns difficult to use. There is, at present, no widely available digital marketplace that supports Telugu-vernacular, bargaining-centred, trust-building livestock trade for this user group.

1.3 Aim and Objectives

The aim of this research was to design and evaluate an inclusive digital marketplace for livestock trading that preserves bargaining practices while improving accessibility, transparency and trust for buyers and sellers in Southern Telangana.

The specific objectives were to:

- Understand current livestock-trading practices, pain points and technology readiness among buyers and sellers in the Nagarkurnool, Wanaparthy and Mahabubnagar districts of Southern Telangana.
- Identify the features and interaction patterns that would increase buyers' and sellers' trust in a digital livestock-trading application.

- Design a mobile application prototype (Sukraven) that supports vernacular-language interaction, digital bargaining and trust-building mechanisms suited to low-literacy rural users.
- Evaluate the usability, trust perception and user satisfaction associated with the resulting prototype among independent buyers and sellers.

1.4 Scope of the Study

The study is limited to the design and evaluation of a mobile application prototype for buyer- and seller-side livestock trading; it does not extend to backend infrastructure, payment-gateway integration, logistics operations or regulatory compliance for a production deployment. The field study and usability evaluation were both conducted with participants drawn from the same three districts of Southern Telangana, and the findings should be interpreted with this regional and linguistic context in mind. Cattle, buffalo, sheep and goats were the livestock categories considered, in line with the animals most commonly traded at the Santhas studied.

1.5 Literature Review

1.5.1 Digital Marketplaces for Agriculture and Livestock in India

The Government of India's electronic National Agricultural Market (e-NAM) has been the principal national effort to modernise agricultural commodity trading through a unified online auction platform. A recent study of farmer awareness and adoption of e-NAM across South Indian states found that while a majority of surveyed farmers possessed basic smartphone literacy, awareness of the platform itself remained around 56 per cent, and that knowledge compatibility with using applications, perceived risk in adoption and participation in awareness programmes were the strongest predictors of awareness, while demographic factors such as age and gender showed no statistically significant association (Manogna, 2025). This is consistent with the position that digital agricultural marketplaces succeed less because of infrastructure alone than because of how well their interaction design matches farmers' existing digital literacy and trust concerns.

A review of the livestock-trading applications most commonly available to Indian farmers, including Pashuwala, Pashushala, Animall and e-NAM, shows a consistent pattern: sellers create a fixed-price listing with photographs and a description, and buyers browse or search listings and then contact the seller directly by phone or WhatsApp to arrange a sale (Pashuwala, 2026; Pashushala, 2026). Price negotiation, where it happens at all, takes place outside the application in an unstructured phone call rather than as a feature of the product itself; an

independent design review of Animall, the most widely adopted of these platforms with reported reach across several North Indian states, observed that in-app bargaining is one of the Indian trading behaviours that such apps have not yet designed for, and that Animall instead only signals informally which sellers are open to negotiation (The Indian Notes, 2023). e-NAM, the Government of India's principal digital agricultural marketplace, follows a unified online auction model rather than a bargaining model, and is not designed around livestock as a product category; a study of e-NAM adoption among South Indian farmers found that awareness of the platform, rather than its fixed-price format, was the main barrier to use (Manogna, 2025). None of these platforms builds structured, in-app negotiation, verified four-angle photographic proof, or a partial-payment reservation mechanism into the transaction itself; each requires a buyer and seller to leave the application, and to establish trust, at the point where the design of the Sukraven prototype (Chapter 5) instead keeps them within it. This pattern under-serves markets, such as the Southern Telangana Santhas studied here, in which price discovery through direct negotiation is not an inefficiency to be removed but a valued and trusted part of the transaction.

1.5.2 Human-Centred Design and Inclusive Design for Rural and Low-Literacy Users

Human-centred design (HCD), as codified in ISO 9241-210, frames interactive-system design as an iterative process grounded in understanding users, their tasks and their environments, rather than in technology-first assumptions (International Organization for Standardization, 2019). Participatory and co-design approaches extend this by treating the people who will use a system as active contributors to its design rather than passive subjects of research (Sanders & Stappers, 2008); this thesis adopted a comparable stance by grounding every subsequent design decision in the field-study findings reported in Chapter 3.

A substantial body of HCI research has examined interface design for low-literacy and novice smartphone users, most notably the ethnographic and experimental work of Medhi and colleagues across India, Kenya, the Philippines and South Africa, which found that non-text, graphical and icon-based interfaces consistently outperformed text-based equivalents for low-literacy participants completing mobile-banking and data-entry tasks (Medhi et al., 2011). These findings motivated the icon-led, image-first interaction patterns used throughout the Sukraven prototype, discussed in Chapter 5.

1.5.3 Trust in Digital Marketplaces

Trust has long been recognised as a central determinant of whether users engage with an unfamiliar online vendor or platform, particularly where familiarity with the counterparty is otherwise low; Gefen (2000) showed, in the context of e-commerce more broadly, that both a consumer's general disposition to trust and their familiarity with the platform's procedures significantly influenced their intention to purchase. In a livestock-trading context, where the 'product' is a living animal whose health and quality cannot be fully verified without physical inspection, and where a buyer and seller may be complete strangers, trust-building mechanisms take on particular importance. This motivated the Sukraven design's emphasis on verified-seller badges, compulsory four-angle photographs and video, seller sales history and ratings, and a partial-payment (Bayana) reservation mechanism, each of which is examined against participant trust ratings in Chapters 3 and 4.

1.5.4 Usability Evaluation

The System Usability Scale (SUS), originally developed by Brooke as a 'quick and dirty' ten-item usability questionnaire (Brooke, 1996), remains one of the most widely used standardised instruments for evaluating the perceived usability of a system. Bangor, Kortum and Miller's large-scale empirical evaluation of the SUS established benchmark interpretations for SUS scores, situating scores above roughly 68 as above average and scores in the 80s as reflecting 'good' to 'excellent' usability (Bangor et al., 2008). This thesis used the SUS, alongside purpose-built trust-perception and user-satisfaction scales, to evaluate the Sukraven prototype, as reported in Chapter 3.

Chapter 2 Methodology

2.1 Research Approach

This research followed a mixed-methods, human-centred design (HCD) approach comprising three phases: (1) a field study combining direct market observation and a structured questionnaire survey to understand current livestock-trading practices and technology readiness; (2) a design and prototyping phase in which the Sukraven mobile application was developed on the basis of the field-study findings; and (3) a usability evaluation of the resulting high-fidelity prototype using validated quantitative instruments. This structure follows the standard HCD cycle of research, synthesis, ideation, prototyping and evaluation (International Organization for Standardization, 2019).

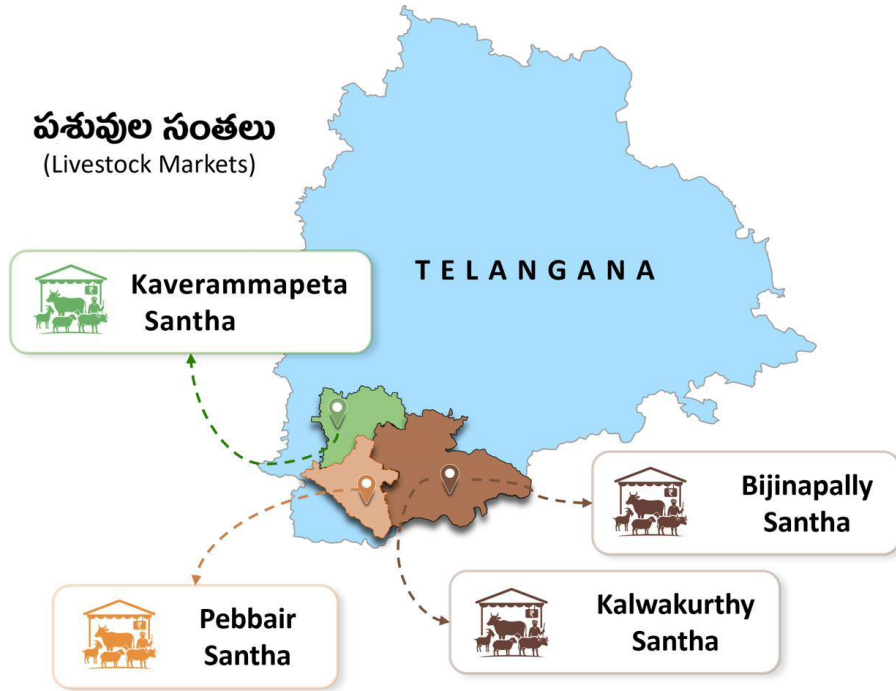


Figure 2.1: Study area — Santhas across the three districts.

2.2 Study Area

The field study was conducted across three districts of Southern Telangana — Nagarkurnool, Wanaparthy and Mahabubnagar — chosen for their concentration of active weekly livestock markets. Figure 2.1 shows the districts and the principal Santhas referenced during the study, including the Kaverammappeta, Pebbair, Bijinapally and Kalwakurthy Santhas. These markets operate on fixed weekly days and draw buyers and sellers from a radius of surrounding villages.



Figure 2.2: Typical layout of a Santha - livestock market.

Figure 2.2 illustrates the typical physical layout of a Santha, based on direct observation. Livestock are grouped by category into a cattle/buffalo trading area and a separate sheep/goat trading area, with a designated parking zone, a small row of shops for food and livestock gear, and a fee-collection point near the single entry/exit route. This layout was used to structure the direct-observation component of the field study described in Section 2.3.1.



Figure 2.3: Bijinapally Santha in Nagarkurnool district.



Figure 2.4: Pebbair Santha in Wanaparthy district.



Figure 2.5: Kaverammampeta Santha in Mahabubnagar district.

2.3 Phase 1: Field Study

The field study was conducted in December 2025 with 82 participants (47 buyers, 35 sellers) recruited through purposive sampling directly at the Santhas in the three study districts, with preference given to buyers and sellers with recent transaction experience. Data were collected

through two complementary methods: direct observation and activity analysis, and a structured, orally-administered questionnaire.

2.3.1 Direct Observation and Activity Analysis

Direct observation and activity analysis (Drury, 1997) was used to record the physical movement patterns of buyers and sellers within the market, the sequence of actions involved in inspecting, bargaining over and completing a livestock purchase, and the points at which delays, confusion or disputes commonly arose. Observation was carried out across multiple market days and informed both the questionnaire design and the buyer/seller journey maps presented in Chapter 3.

LIVESTOCK MARKET PROCESS

The full transaction flow I observed

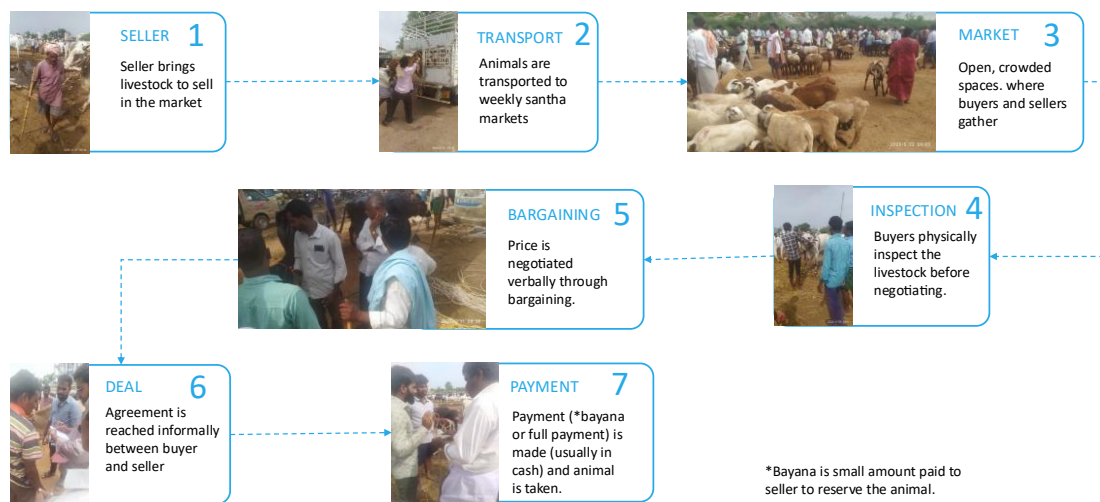


Figure 2.6: Livestock market transaction process.

2.3.2 Questionnaire

A structured questionnaire was developed for this study and administered orally in Telugu to accommodate participants with limited reading ability, with responses recorded by the researcher. The questionnaire was organised into four sections: Section A recorded participant profile (role, gender, age group, education level, primary livestock traded, years of trading experience and frequency of visiting the weekly Santha); Section B covered market experience specific to sellers (questions 8–12), buyers (questions 13–16) and both groups in common (questions 17–20), including price determination, challenges faced, satisfaction with outcomes, expenses incurred, exposure to unfair practices, and factors that build trust between trading parties; Section C assessed technology awareness and digital readiness (questions 21–28),

including phone ownership and usage, prior experience with online trading platforms, willingness to use a mobile application, anticipated difficulties, and the features that would increase trust in and shape expectations of such an application; and Section D captured agreement, on a five-point scale from Strongly Disagree to Strongly Agree, with eight perception statements (questions 29–36) concerning market difficulty, the importance of bargaining, confidence in assessing quality, satisfaction with the current process, comfort with smartphones and receptiveness to a digital marketplace. The full questionnaire, in English and in the Telugu form used in the field, is reproduced in Appendix A and Appendix B respectively. Free-text and multiple-selection responses (Sections B and C) were manually coded into categories; where two response labels described the same underlying category (for example, near-identical phrasings such as "transportation difficulties" and "transportation problems", or "low price" and "receiving a lower price than expected"), these were merged into a single category and the combined count is reported in Chapter 3. Where a participant's response was recorded against more than one category (for example, a participant who named both livestock photographs and videos), that response is counted once within each relevant category, so category totals for a given question do not necessarily sum to 82. This coding, along with the thematic and affinity analysis of interview and observation notes (Section 2.3.3), was carried out by a single researcher, who was also the designer of the Sukraven prototype and the sole moderator and oral administrator of every questionnaire, interview and usability session reported in this thesis. No second coder cross-checked the categorisation, and no inter-coder reliability statistic was computed; this single-researcher design is acknowledged as a reflexivity and bias limitation in Section 6.2, alongside the related risk that participants responding face-to-face to the same person who built the application may have rated it more favourably out of politeness than they would have to an independent evaluator.

Table 2.1: Participants across the two phases of the study

Phase	Purpose	Buyers	Sellers	Total	Timing
Field study	Understand practices, pain points and technology readiness	47	35	82	December 2025
Usability evaluation	Evaluate the Sukraven prototype	10	8	18	May 2026

2.3.3 Interviews and Thematic Analysis

Alongside the questionnaire, structured face-to-face interviews of 15 to 25 minutes were conducted in Telugu with a subset of buyers and sellers at the Santhas, exploring bargaining practices, trust concerns and prior experience with intermediaries in greater depth than the

questionnaire's fixed-response format allowed. Interview notes and observation field notes were analysed using thematic and affinity analysis: statements and observations were transcribed, grouped into recurring themes, and clustered on an affinity diagram to identify the design opportunity areas that motivated the Sukraven concept (Chapter 5). This qualitative analysis, together with the observation record (Section 2.3.1) and the questionnaire findings (Section 2.3.2), fed into the synthesis described in Section 2.4. The exact number of participants interviewed and the specific procedure used to select this subset from the 82 field-study participants were not separately logged at the time of data collection and could not be reconstructed from the surviving field notes for this thesis; this is acknowledged as a methodological gap in Section 6.2 rather than presented as a settled figure. The three Telugu-language instruments used in this study (the field-study questionnaire and, in Phase 3, the SUS, Trust Perception and User Satisfaction scales) were translated from their English originals by the researcher for oral administration but were not independently back-translated or otherwise formally validated prior to use; this is a limitation that bears directly on the interpretation of the reverse-worded SUS item responses discussed in Section 6.2.

2.4 Phase 2: Design and Prototyping

Findings from the field study were synthesised into design requirements and used to inform the information architecture, user flows, visual design system and interaction patterns of the Sukraven prototype, described fully in Chapter 5. The prototype was developed iteratively from low-fidelity wireframes to a high-fidelity, bilingual (Telugu and English) interactive prototype, covering onboarding and authentication, home browsing and search, livestock listing detail, bargaining and Bayana reservation, payment, delivery coordination, listing creation and account/profile management.

2.5 Phase 3: Usability Evaluation

The high-fidelity Sukraven prototype was evaluated in May 2026 with 18 independent participants (10 buyers, 8 sellers) who had not taken part in the December 2025 field study. These 18 participants were recruited using the same purposive-sampling approach as the field study: buyers and sellers were approached directly at the weekly Santhas in the Nagarkurnool, Wanaparthy and Mahabubnagar districts, with preference given to those with recent transaction experience, drawn from the same market population as the 82 field-study participants but not overlapping with them. Each participant completed a moderated session interacting with the prototype's core buyer or seller tasks, followed by three validated questionnaires: the System

Usability Scale, a Trust Perception Scale, and a User Satisfaction Scale, each administered orally in Telugu by the researcher, in the same manner as the field-study questionnaire (Section 2.3.2), using Telugu-language versions of all three instruments. All items, including the five negatively-worded SUS items (SUS2, 4, 6, 8, 10), were read to participants exactly as translated, without rewording into a positive form, facilitator restatement of the item's intent, or a supplementary visual response aid.

2.5.1 System Usability Scale (SUS)

Following standard usability-engineering practice of pairing task-based observation with a standardised post-task instrument (Nielsen, 1994), the System Usability Scale (Brooke, 1996) is a widely used ten-item, five-point Likert questionnaire yielding a single usability score between 0 and 100. Odd-numbered items (SUS1, 3, 5, 7, 9) are scored as (response – 1), and even-numbered items (SUS2, 4, 6, 8, 10) are scored as (5 – response); the sum of these contributions is multiplied by 2.5 to yield the final score. Bangor, Kortum and Miller's benchmark interpretation situates scores above 80.3 as 'Excellent', 68–80.3 as 'Good', around 68 as 'Average', 51–67 as 'OK/Marginal', and below 51 as 'Poor' (Bangor et al., 2008).

2.5.2 Trust Perception Scale

A six-item Trust Perception Scale was developed for this study to assess the extent to which specific Sukraven design features increased participants' trust in the platform, using a five-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree). The items addressed verified-profile and health badges (T1), four-angle photo and video proof (T2), Bayana deposit security (T3), bargain transparency (T4), seller ratings and sales history (T5), and localised Telugu-language interface (T6).

2.5.3 User Satisfaction Scale

A five-item User Satisfaction Scale was developed to assess overall satisfaction (S1), perceived efficiency relative to visiting a physical Santha (S2), fulfilment of expectations (S3), likelihood of recommending the application to other traders (S4), and perceived value in obtaining fair, transparent prices (S5), again using a five-point Likert scale.

2.6 Ethical Considerations

Consent for both the December 2025 field study and the May 2026 usability evaluation was obtained orally in Telugu, immediately before each session, in the field setting; the purpose of the research and the intended use of responses were explained to each participant by the

researcher before proceeding. Written or signed consent forms were not used for either phase, and formal institutional ethics clearance had not been obtained prior to data collection. An Ethical Clearance Form describing the study in full, including this limitation, was subsequently submitted to the PDPM IIITDM Jabalpur Institutional Ethics Committee (IEC) on 29 July 2026 (institutional Ethical Clearance Form, Annexure-I); as of the submission of this thesis, the IEC had not yet issued a decision on this application, and no approval reference number had been assigned. The ethical status of the full dataset (82 field-study and 18 usability-evaluation participants) therefore remains open pending that outcome, and this is acknowledged, not as a closed matter, but as an unresolved item to be settled before the thesis defence (Section 6.2). The consent script actually read to participants before each session, covering the purpose of the research, the voluntary nature of participation, the right to withdraw without penalty, and how responses would be used and stored, is reproduced in Appendix C. Usability-evaluation records used participant identifiers and role only, with no names retained in the analysis reported in this thesis.

Chapter 3 Results

This chapter reports findings from the two data-collection phases described in Chapter 2: the December 2025 field study (n = 82) and the May 2026 usability evaluation of the Sukraven prototype (n = 18). Results are presented in the same sequence as the methodology.

3.1 Field Study: Direct Observation and Activity Analysis

Direct observation across the study Santhas confirmed the physical layout summarised in Figure 2.2 and surfaced several recurring patterns in buyer and seller behaviour. Livestock were grouped informally by category (cattle/buffalo and sheep/goats), and buyers moved between groups comparing multiple animals before approaching a seller to bargain. Bargaining itself proceeded through an iterative sequence of offer and counter-offer conducted verbally and in person, typically concluding with a handshake or verbal agreement rather than any written record. No formal price board or reference pricing was observed at any of the markets visited, consistent with participants' own report that no fixed-pricing system exists in their local market (see Section 3.4). Transport of purchased animals, either by the buyer or arranged informally with a transporter present at the market, was observed as a recurring source of delay and negotiation after the sale itself was agreed.

3.2 Field Study: Participant Profile

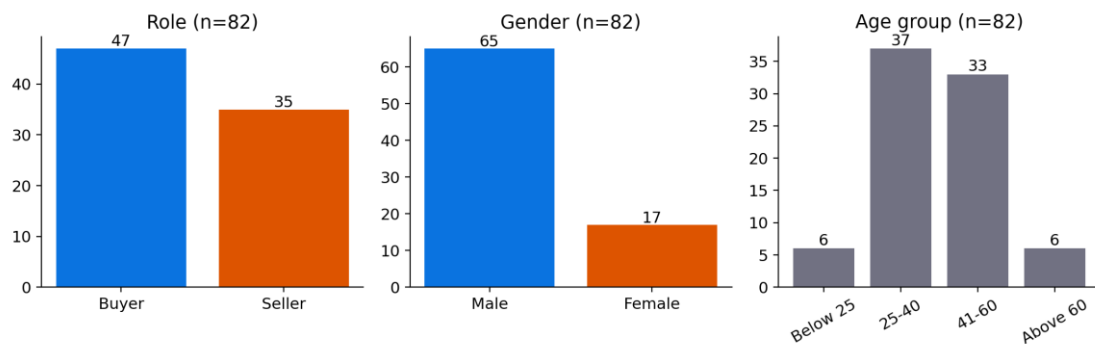


Figure 3.1: Participant profile (role, gender, age).

Table 3.1: Demographic profile of field-study participants (n = 82)

Attribute	Category	n	%
Role	Buyer	47	57.3
	Seller	35	42.7
Gender	Male	65	79.3
	Female	17	20.7

Age group	Below 25 years	6	7.3
	25–40 years	37	45.1
	41–60 years	33	40.2
	Above 60 years	6	7.3
Education	No formal education	24	29.3
	Primary	16	19.5
	Secondary	22	26.8
	Higher secondary	7	8.5
	Graduate and above	13	15.9
Primary livestock traded	Sheep	26	31.7
	Mixed	17	20.7
	Goat	16	19.5
	Buffalo	12	14.6
	Cow	11	13.4
Frequency of visiting Santha	Every week	38	46.3
	Occasionally	23	28.0
	Twice a month	21	25.6

Almost half of participants (45.1%) fell in the 25–40 age band, and a further 40.2% were aged 41–60, indicating that the majority of active traders were of working age rather than elderly. Nearly 30% of participants reported no formal education, and a further 19.5% had only primary-level education, underlining the importance of low-literacy-friendly, vernacular interface design for any digital solution aimed at this population (Medhi et al., 2011). Sheep and mixed-livestock trading accounted for over half of participants' primary trade (31.7% and 20.7% respectively), reflecting the composition of the Santhas visited.

3.3 Field Study: Market Experience

3.3.1 Selling Experience

Among the 35 sellers surveyed, price was most commonly determined through direct negotiation with buyers rather than any external reference; sellers described basing their asking price on physical appearance, weight, age and prevailing informal market rates. Only 12 of the 35 sellers (34.3%) reported being satisfied with the final selling price they received, while 23 (65.7%) were not; the most frequently cited reason for dissatisfaction was receiving a lower price than expected, often attributed to weak bargaining position, a lack of alternative buyers on the day, or broker interference. Transportation was the most commonly reported expense (29 of 35 sellers), followed by market entry fees (15) and animal feeding costs (6). Close to

half of sellers (17 of 35, 48.6%) reported having experienced unfair practices while trading, most commonly involving underpricing or pressure tactics from brokers and intermediaries.

3.3.2 Buying Experience

Among the 47 buyers surveyed, livestock quality was most commonly evaluated through physical inspection of health, weight, age (via teeth examination) and, for dairy animals, milk-yielding capacity. Only 19 of 47 buyers (40.4%) reported feeling confident about livestock quality prior to purchase, while 28 (59.6%) did not, indicating a substantial gap in buyers' ability to verify an animal's condition before committing to a purchase — a gap directly addressed in the Sukraven design through compulsory four-angle photographs, video and health-detail fields (see Chapter 5). Price finalisation for buyers occurred predominantly through direct bargaining (35 of 47) or mutual discussion (12 of 47); no buyer reported relying on a fixed price.

Table 3.2: Most frequently reported challenges in selling and buying livestock

Sellers' challenges (n = 35)	n	Buyers' challenges (n = 47)	n
Receiving a lower price than expected	23	Difficulty assessing livestock quality	14
Difficulty finding genuine buyers	12	Unclear pricing	13
Transportation difficulties	8	Trust concerns	11
Animal health concerns	6	Difficulty finding suitable livestock	7
		Transportation difficulties / problems	11

3.3.3 Bargaining, Pricing and Trust

All 82 participants (100%) reported that there is no fixed-pricing system in their local livestock market, confirming that price discovery through negotiation is the universal norm rather than an occasional practice. Sixty-three participants (76.8%) rated bargaining as 'Very Important' to obtaining a fair price, and the remaining 19 (23.2%) rated it 'Moderately Important'; not a single participant considered bargaining unimportant. When asked what factors help them trust the other party during a transaction, participants most frequently cited previous transactions with the same person (17 mentions), general honesty (15), face-to-face familiarity (15) and transparency of information (13), with a further six participants each citing being known people or being from the same village. This pattern — trust grounded overwhelmingly in repeat interaction and personal familiarity rather than any formal verification mechanism —

motivated the emphasis on verified profiles, seller sales history and ratings in the Sukraven design (Chapter 5; Gefen, 2000).

Asked what improvements they would like to see in the current livestock-trading system, participants most frequently mentioned better transportation facilities (19 mentions), greater transparency (14), proper price information (11), a transparent pricing system specifically (6), reduced broker control or elimination of brokers (18 combined), and digital payments or platforms (12 combined); combining the three pricing- and transparency-related answers (transparency, proper price information and transparent pricing system) gives 31 of 82 participants (37.8%) volunteering a pricing-transparency theme when asked what they wanted improved, even though none selected transparent pricing from the closed list of trust features in Section 3.4.

3.4 Field Study: Technology Awareness and Digital Readiness

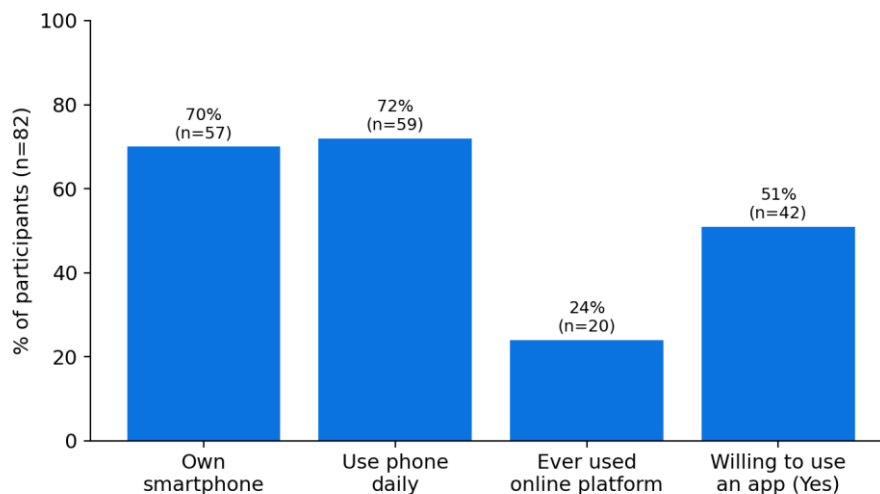


Figure 3.2: Mobile technology ownership and usage.

A majority of participants owned a smartphone (57 of 82, 69.5%), while the remainder owned only a basic phone (25 of 82, 30.5%); no participant reported owning no phone at all. Fifty-nine participants (72.0%) used their phone daily, and mobile phones were used overwhelmingly for calls (81 of 82) and WhatsApp (41 of 82), with UPI payments already in use by 29 participants (35.4%) — indicating meaningful digital-payment familiarity even where dedicated trading-platform usage was low. Only 20 of 82 participants (24.4%) had ever used any online platform to buy or sell livestock, yet 42 (51.2%) said they would be willing to use a mobile application for this purpose, with a further 17 (20.7%) saying 'Maybe' and 23 (28.0%) saying 'No'. The most frequently anticipated difficulties in using such an application

were a lack of smartphone knowledge (27 mentions), reading difficulty (19), trust issues (17), language (14 mentions, combined with a further 5 citing 'language problem' specifically), and internet connectivity (12 mentions, combining 11 citing 'internet connectivity' and 1 citing 'internet issue').

When asked which features would increase their trust in a livestock-trading application, participants most frequently selected verified user profiles (45 combined mentions), livestock videos (22), livestock photographs (21) and health and vaccination details (6); five participants who named both photographs and videos together are counted in both totals. User ratings and reviews were named by only two participants, and no participant named transparent pricing as a trust-building feature specifically, while eleven participants said they did not know. These last two results are reported for completeness: verified profiles and visual proof (photographs and video) are strongly and directly supported by this finding, whereas seller ratings are not supported by any question in the field study. Pricing transparency is a partial exception: although it drew no mentions on this particular closed-ended trust-features question, a related but separate question asking what improvements participants wanted to see in the current trading system (Section 3.3.3) drew pricing- and transparency-related answers from 31 of 82 participants, so the field study does provide some indirect support for pricing transparency as a design priority, even though it was not selected as a trust feature per se; the design rationale in Chapter 5 treats this distinction explicitly. When asked which features they would expect in such an application, livestock details such as breed, age, weight and health (37 mentions) and price guidance (18) were the most commonly named, ahead of digital bargaining (7), search and filters (5) and distance to seller (5); notably, 22 participants said they did not know what to expect, reflecting limited prior exposure to comparable digital tools.

Table 3.3: Features participants said would increase trust in a livestock-trading application (n = 82, multiple selection)

Feature	Mentions
Verified user profiles	45
Livestock videos	22
Livestock photographs	21
Health and vaccination details	6
User ratings and reviews	2
Transparent pricing	0
Did not know / no answer	11

3.5 Field Study: Perception Statements

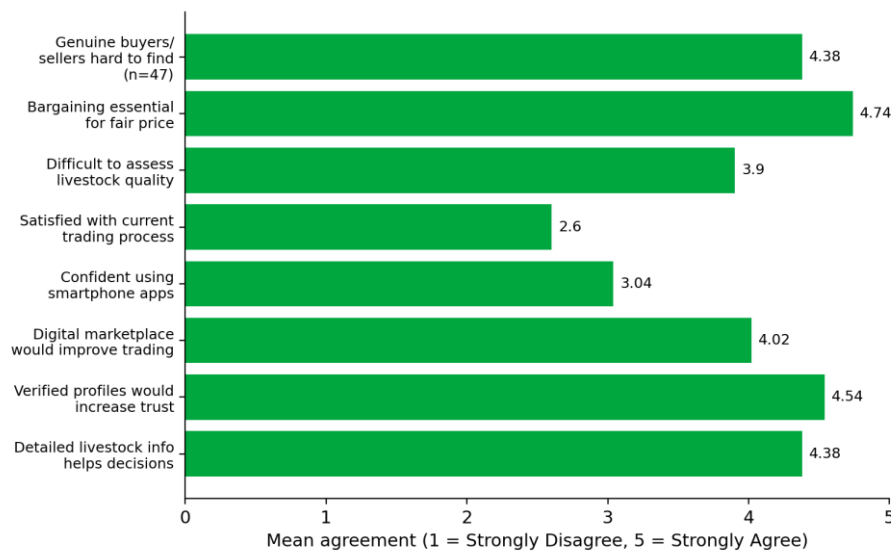


Figure 3.3: Mean agreement with perception statements.

Participants strongly agreed that bargaining is essential for a fair price ($M = 4.74$, $SD = 0.44$) and that verified profiles would increase their trust in a trading application ($M = 4.54$, $SD = 0.74$). Across all 82 participants, finding genuine buyers or sellers was rated as difficult ($M = 4.40$, $SD = 0.49$). Detailed livestock information was also strongly endorsed as helpful for decision-making ($M = 4.38$, $SD = 0.73$), and difficulty assessing livestock quality before purchase was rated moderately high ($M = 3.90$, $SD = 0.66$). By contrast, satisfaction with the current trading process was the lowest-rated item overall ($M = 2.60$, $SD = 0.73$), indicating clear dissatisfaction with the status quo, while confidence using smartphone applications was rated only moderately ($M = 3.04$, $SD = 1.59$), reflecting a wide spread of digital comfort across the sample. A majority nonetheless agreed that a digital marketplace would improve livestock trading ($M = 4.02$, $SD = 1.09$). Read together, these results indicate a population that is dissatisfied with current trading conditions, values bargaining highly, and is cautiously receptive to a well-designed digital alternative, provided such an alternative retains negotiation and demonstrable trust signals rather than imposing a fixed-price, unfamiliar interaction model.

3.6 Buyer and Seller Journey Maps

Observation and interview findings were consolidated into buyer and seller journey maps spanning need recognition through to returning home, identifying the emotional highs and lows and pain points at each stage, and the corresponding design opportunity for Sukraven.

Table 3.4: Buyer journey map: stages, pain points and design opportunities

Stage	Emotion	Pain point	Design opportunity
Need recognition	Hopeful	Limited market information	Digital marketplace
Travel to Santha	Tired	Long travel time and cost	Remote browsing
Search & inspect	Uncertain	Difficult to compare many animals	Rich livestock information
Negotiate	Stressed	Unclear pricing and trust concerns	Digital bargaining
Purchase	Relieved	Risk of paying an unfair price	Transparent pricing
Transport home	Satisfied	Transportation difficulties	Delivery coordination

Table 3.5: Seller journey map: stages, pain points and design opportunities

Stage	Emotion	Pain point	Design opportunity
Prepare livestock	Optimistic	Time spent preparing	Digital listings
Travel to Santha	Tired	High transportation costs	Eliminate unnecessary travel
Display livestock	Anxious	Limited buyer reach	Reach more buyers online
Negotiate	Frustrated	Low offers and repeated negotiations	Digital bargaining
Sell	Happy / disappointed	Unsold livestock	Secure advance booking (Bayana)
Return home	Relieved	Additional return costs	Reduced transport burden

3.7 Usability Evaluation Results

The Sukraven prototype was evaluated with 18 independent participants (10 buyers, 8 sellers) using the System Usability Scale, the Trust Perception Scale and the User Satisfaction Scale described in Section 2.5.

3.7.1 System Usability Scale

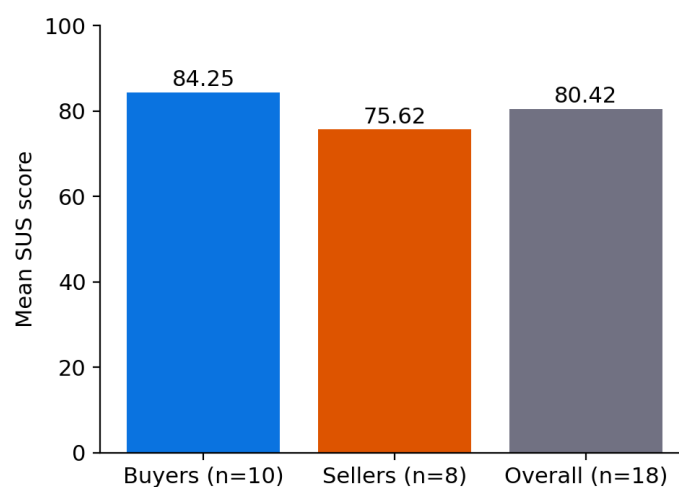


Figure 3.4: Mean SUS score by participant role.

Table 3.6: System Usability Scale scores by participant (n = 18)

Participant	Role	SUS Score
U01	Buyer	87.5
U02	Buyer	80.0
U03	Buyer	87.5
U04	Buyer	85.0
U05	Buyer	75.0
U06	Buyer	92.5
U07	Buyer	82.5
U08	Buyer	85.0
U09	Buyer	82.5
U10	Buyer	85.0
U11	Seller	72.5
U12	Seller	75.0
U13	Seller	80.0
U14	Seller	67.5
U15	Seller	82.5
U16	Seller	77.5
U17	Seller	80.0
U18	Seller	70.0

To address the response pattern on the reverse-worded items directly, Table 3.6a reproduces the raw item-level responses for the five negatively-worded SUS statements (SUS2, 4, 6, 8, 10) for all 18 participants, on the original 1–5 scale (1 = Strongly Disagree, 5 = Strongly Agree).

Table 3.6a: Raw responses to the five reverse-worded SUS items, by participant (1–5 scale, n = 18)

Participant	Role	SUS2	SUS4	SUS6	SUS8	SUS10
U01	Buyer	2	1	2	2	1
U02	Buyer	2	2	1	2	2
U03	Buyer	1	2	2	1	2
U04	Buyer	2	1	2	2	1
U05	Buyer	2	2	2	2	2
U06	Buyer	1	1	2	2	1
U07	Buyer	2	2	1	2	1
U08	Buyer	2	2	2	1	2
U09	Buyer	1	2	2	2	1
U10	Buyer	2	2	2	2	2
U11	Seller	2	3	2	2	2
U12	Seller	2	2	2	3	2
U13	Seller	2	2	2	2	1
U14	Seller	3	3	2	3	2
U15	Seller	2	2	2	2	2
U16	Seller	2	2	1	2	2
U17	Seller	2	2	2	2	2
U18	Seller	2	3	2	2	3

No participant selected 'Agree' or 'Strongly Agree' (4 or 5) on any of the 90 reverse-worded-item responses in this table, confirming the polarity-tracking result reported above. The responses are not, however, a uniform single value: they range across 'Strongly Disagree' (1, 17 of 90 responses), 'Disagree' (2, 66 of 90) and 'Neutral' (3, 7 of 90), with the small cluster of Neutral responses concentrated among three sellers (U11, U12, U14, U18) who also recorded comparatively lower overall SUS scores (Table 3.6). This graded variation is more consistent with participants engaging differentially with each reversed statement than with a single undifferentiated 'disagree with everything negative' response strategy, and to that extent it offers some reassurance against the concern that the reversed items were not understood at all. This thesis nonetheless stops short of treating this pattern as confirmed evidence of comprehension: no session recordings or cognitive-interview probes of the reversed items were conducted (Section 6.2), so an oral-delivery or translation artefact affecting all five reverse items in the same direction cannot be ruled out from this data alone. Accordingly, the headline SUS figure of 80.42 is reported throughout this thesis, including in the Abstract, Chapter 4 and Chapter 6, together with this caveat, and the reader is directed to Section 6.2 for the full discussion of what would be needed to resolve the question conclusively.

The overall mean SUS score was 80.42 (SD = 6.54, n = 18), which falls within the 'Good' usability band, not at the boundary with 'Excellent' on Bangor, Kortum and Miller's benchmark scale; given the standard deviation of 6.54 at n = 18, the confidence interval around this mean does technically straddle the Good/Excellent seam at approximately 80.3, but this thesis reports the conservative 'Good' classification throughout, consistent with Section 4.3 and Section 6.1, rather than the more favourable 'Excellent' or letter-grade readings, and this conservative reading should be read together with the validity caveat on the reverse-worded SUS items discussed immediately below and in Section 6.2. Buyers scored the prototype somewhat higher on average (M = 84.25) than sellers (M = 75.62); the lowest individual score recorded was 67.5 (participant U14, a seller), and the highest was 92.5 (participant U06, a buyer). No participant scored below the 'OK/Marginal' threshold of 51, indicating that even the least favourable individual response reflected acceptable, rather than poor, usability.

3.7.2 Trust Perception Scale

Table 3.7: Trust Perception Scale item means (1–5 scale, n = 18)

Item	Description	Mean	SD
------	-------------	------	----

T1	Verified profile & health badges increase confidence	4.44	0.51
T2	Four-angle photos & videos help assess quality before a deal	4.50	0.51
T3	Bayana deposit system provides financial security	4.28	0.57
T4	Bargain chat provides clear, transparent pricing	4.06	0.64
T5	Seller ratings & sales history build trust in the counterparty	4.11	0.58
T6	Telangana Telugu interface feels safe and relatable	4.44	0.51

The overall mean Trust Perception score was 4.31 out of 5. The four-angle photo and video requirement (T2, M = 4.50) and localised Telugu-language interface (T6, M = 4.44) were rated as the strongest contributors to trust, alongside verified profile and health badges (T1, M = 4.44). Bargain transparency (T4, M = 4.06) was rated comparatively lower, though still solidly on the 'Agree' side of the scale, suggesting that while the offer/counter-offer chat interface was seen positively, there remains some room to make negotiated pricing feel more transparent.

3.7.3 User Satisfaction Scale

Table 3.8: User Satisfaction Scale item means (1–5 scale, n = 18)

Item	Description	Mean	SD
S1	Overall satisfaction with the Sukraven experience	4.33	0.59
S2	Efficiency compared with visiting a weekly Santha	4.39	0.61
S3	The app met expectations for buying/selling livestock online	4.06	0.54
S4	Would recommend Sukraven to other buyers/sellers	4.44	0.51
S5	Sukraven can help obtain fair, transparent prices	4.22	0.55

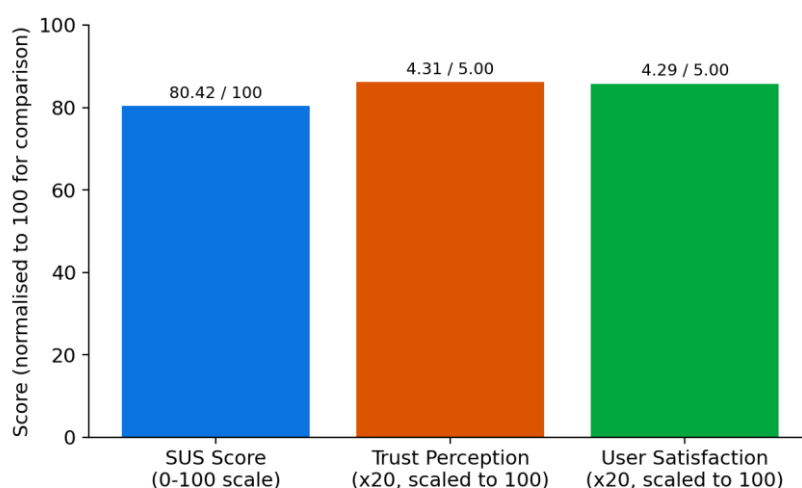


Figure 3.5: Summary of usability-evaluation instruments.

The overall mean User Satisfaction score was 4.29 out of 5. Participants were most willing to recommend Sukraven to other buyers and sellers (S4, M = 4.44) and rated its efficiency relative

to a physical Santha visit highly (S2, $M = 4.39$). Expectation fulfilment (S3, $M = 4.06$) was the comparatively lowest-rated item, though it remained on the 'Agree' side of the scale, suggesting that while the prototype broadly met participants' expectations, some gap between anticipated and delivered functionality remains, discussed further in Chapter 4.

Chapter 4 Discussion

This chapter interprets the results reported in Chapter 3 with reference to the literature reviewed in Chapter 1, and follows the same sequence: field-study findings on trading practice and technology readiness, followed by the usability-evaluation findings for the Sukraven prototype.

4.1 Bargaining as a Design Requirement, Not a Design Problem

The field study found unanimous agreement (100% of 82 participants) that no fixed-pricing system exists in the study markets, and that bargaining is regarded as 'Very Important' or 'Moderately Important' to obtaining a fair price by every single participant. This finding directly challenges the fixed-listing, catalogue-style structure adopted by many existing agricultural and livestock e-commerce platforms. Where general e-commerce research treats negotiation as friction to be removed, the results here indicate that, for this population, bargaining is itself the mechanism through which fairness and trust are established. This is consistent with the low satisfaction reported by sellers with their final selling price (34.3% satisfied) despite bargaining being universally practised: dissatisfaction appears to stem less from the presence of negotiation than from information asymmetry, weak bargaining position and broker interference, which a well-designed digital bargaining flow can help to address by giving both parties visibility of the offer history and a documented final agreed price. This reasoning directly informed Sukraven's decision to retain an explicit offer/counter-offer chat interface (Chapter 5) rather than adopting fixed pricing, as seen in comparable Indian livestock platforms.

4.2 Trust Is Built Through Familiarity, and Must Be Rebuilt Digitally

Participants' own account of what builds trust between trading parties -- overwhelmingly previous transactions, honesty and face-to-face familiarity rather than any formal credential -- echoes Gefen's (2000) finding that both disposition to trust and familiarity with a platform's procedures shape a user's willingness to transact. A purely digital marketplace necessarily removes the face-to-face familiarity that underpins trust in the physical Santha; the trust-perception results (Section 3.7.2) suggest that Sukraven's verified-profile badges, four-angle photo and video requirements, and seller ratings/sales-history display can partially substitute for this lost familiarity, with the two highest-rated trust items being visual proof (T2, M = 4.50) and the localised Telugu interface (T6, M = 4.44). This is consistent with the field-study finding

that buyers' confidence in livestock quality before purchase was low (40.4% confident) and that verified profiles and livestock photographs/videos were the features most frequently named as would increase trust in a digital platform (Section 3.4).

4.3 Designing for Low Formal Education and Variable Smartphone Comfort

Almost 30% of field-study participants reported no formal education, and confidence using smartphone applications was rated only moderately in the perception statements ($M = 3.04$, $SD = 1.59$) -- the widest spread of any item, indicating that the study population spans a considerable range of digital comfort rather than being uniformly low- or high-literacy. This is consistent with Medhi et al.'s (2011) finding that non-text, icon- and image-led interfaces substantially outperform text-heavy equivalents for low-literacy and novice smartphone users. Sukraven's reliance on large iconography (Section 5.4), a bilingual Telugu/English interface selectable at first launch, and image-first listing and profile screens (Sections 5.5 to 5.8) reflects this literature directly. The usability-evaluation results are consistent with this design choice: the overall SUS score of 80.42 falls in the 'Good' band (subject to the reverse-worded-item validity caveat discussed in Section 3.7.1 and Section 6.2), and every individual SUS score exceeded the 'OK/Marginal' threshold of 51 (Bangor et al., 2008). This thesis cannot, however, claim on the basis of the 18-participant usability-evaluation sample specifically that this result was achieved with participants who had no formal education: the individual demographic characteristics of the 18 usability-evaluation participants were not separately tabulated (Section 6.2), and this claim should not be read as extending beyond the 82-participant field-study sample in which the no-formal-education figure was actually measured (Section 3.2).

4.4 The Buyer-Seller Usability Gap

Buyers rated the prototype's usability somewhat more favourably than sellers (mean SUS of 84.25 versus 75.62). This gap likely reflects the greater complexity of the seller-side task flow: creating a listing requires navigating a four-step process (photos and video, basic information, health and breeding details, and price and location; see Section 5.8) compared with the comparatively linear buyer flow of browsing, viewing a listing and initiating a bargain. The lowest individual SUS score in the sample (67.5) was recorded by a seller (participant U14), who also recorded the lowest trust and satisfaction scores among all 18 participants, suggesting that for at least a subset of sellers, the listing-creation flow may benefit from further simplification in future iterations.

4.5 Meeting, but Not Exceeding, Expectations

Expectation fulfilment was the lowest-rated of the five User Satisfaction items (S3, M = 4.06), notwithstanding an otherwise strongly positive satisfaction profile, including a high willingness to recommend the application (S4, M = 4.44). Combined with the field-study finding that 22 of 82 participants (26.8%) said they did not know what features to expect from a livestock-trading application (Section 3.4), this suggests that many participants approached the prototype with limited prior reference points, and that the prototype's actual functionality, while well received, did not fully anticipate needs that only became apparent through hands-on use, such as more granular price guidance or a wider range of search filters. This points to an opportunity for continued iterative evaluation as adoption grows, discussed further in Section 6.3.

4.6 Relationship to e-NAM and Broader Digital Agriculture Marketplace Adoption

The pattern observed here -- reasonably high smartphone ownership (69.5%) alongside very low prior use of any online trading platform (24.4%) -- parallels findings from the national e-NAM platform, where awareness and adoption lag behind general digital literacy, and where knowledge compatibility with using applications, rather than age or gender, is the strongest demographic predictor of awareness (Manogna, 2025). This suggests that the barrier to digital livestock-marketplace adoption in Southern Telangana is less a lack of underlying digital capability than a mismatch between existing platforms' interaction design and the specific practices, bargaining, vernacular language, and trust built through familiarity, that define how this population actually trades.

Chapter 5 Concepts - The Sukraven Design

This chapter presents the Sukraven mobile application, the design concept developed from the field-study findings reported in Chapter 3. It follows the structure of the human-centred design process introduced in Section 2.4: information architecture, user flows, the visual design system, and the core interaction flows of the application, each explained with reference to the field-study finding that motivated it.

5.1 Design Process and Rationale

Sukraven was developed through an iterative human-centred design process (International Organization for Standardization, 2019), moving from the field-study synthesis to low-fidelity wireframes and then to a high-fidelity, bilingual (Telugu and English) interactive prototype. Three findings from the field study shaped every subsequent design decision: first, that bargaining is universally practised and highly valued (Section 3.3.3), which ruled out a fixed-price catalogue model in favour of a persistent offer/counter-offer interaction; second, that trust is built through familiarity and direct verification rather than formal credentials (Section 3.3.3), which motivated verified-profile badges and compulsory four-angle photography and video; these two design decisions are directly supported by the field study, in which verified profiles were named by 45 of 82 participants and photographic or video proof by 21 and 22 respectively as features that would increase their trust (Section 3.4). Visible seller ratings and sales history were also included, but this feature is not supported by the field-study trust-features question, on which only two participants named ratings; it is instead a design inference drawn from the trust literature (Gefen, 2000; McKnight, Choudhury & Kacmar, 2002) and from participants' emphasis on repeat dealing with known individuals as a basis for trust (Section 3.3.3), and is reported here as a theoretically motivated rather than an empirically grounded decision. The same standard is applied to two further features. Transparent, itemised bargaining chat (Section 5.7) is treated as a partially, rather than fully, field-supported decision: it drew no mentions on the closed-ended trust-features question (Section 3.4), but 31 of 82 participants separately volunteered a pricing-transparency theme when asked what improvements they wanted to see (Section 3.3.3), so it is reported as an inference reinforced by indirect field evidence rather than as a directly evidenced requirement. The Bayana (token-deposit) reservation mechanism (Section 5.7), by contrast, was not named or implied by any participant in either the trust-features question, the expected-features question, or the open-ended improvements question,

and is reported here as a purely theoretically motivated design inference, drawn from the researcher's observation of informal, unenforced verbal deposits at the Santhas (Section 3.1) and from the general literature on trust and commitment devices in online transactions (Gefen, 2000), rather than as an empirically grounded requirement. And third, that formal education levels and smartphone confidence vary widely across the user base (Section 3.2, 3.4), which motivated an icon-led, low-text interaction style privileging recognition over recall (Norman, 2013) and a Telugu-first language default, consistent with Medhi et al.'s (2011) findings on low-literacy mobile interface design and with established interaction-design practice for reducing cognitive load in novice users (Preece, Rogers & Sharp, 2015).

5.2 Information Architecture

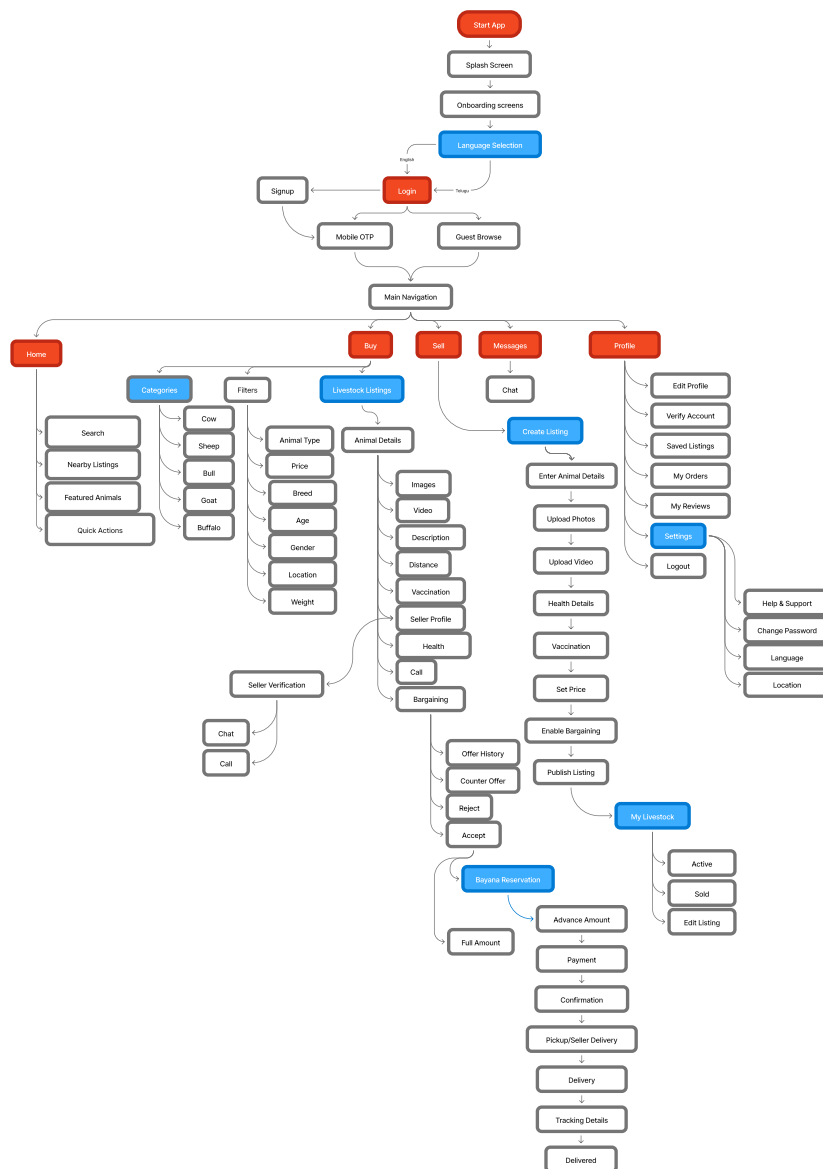


Figure 5.1: Sukraven information architecture.

The information architecture (Figure 5.1) organises the application around five primary navigation destinations reached from a persistent bottom navigation bar: Home, Buy, Sell, Messages and Profile. The Buy branch nests category browsing, filters and livestock listings, each livestock listing in turn exposing seller verification, health information and the bargaining flow. The Sell branch nests the four-step listing-creation flow and the seller's My Livestock dashboard. This flat, category-first structure was chosen so that a first-time user reaches a livestock listing or the start of a sale within two taps of opening the application, reducing the navigational burden on users with limited prior app experience.

5.3 User Flows

5.3.1 Login and Signup

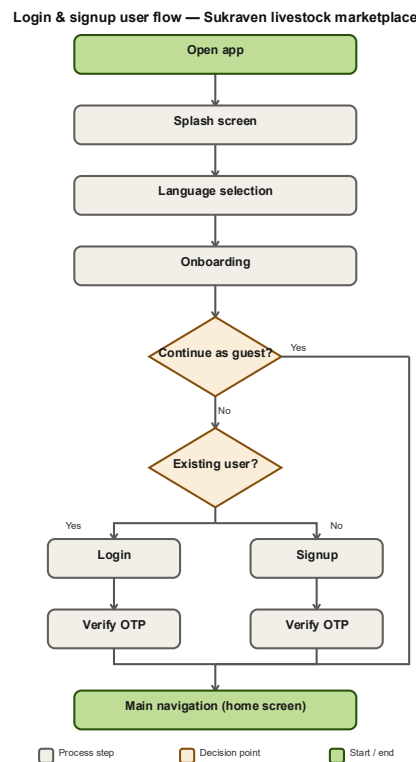


Figure 5.2: Login and signup user flow.

The login and signup flow (Figure 5.2) opens with language selection between Telugu and English before any other content is shown, reflecting the field-study finding that language was one of the most frequently anticipated difficulties in using a trading application (Section 3.4). Users may continue as a guest to browse listings immediately, or authenticate via mobile OTP, deferring account creation until a user is ready to bargain or list an animal, so as not to place a registration barrier in front of first-time exploration.

5.3.2 Buyer Flow

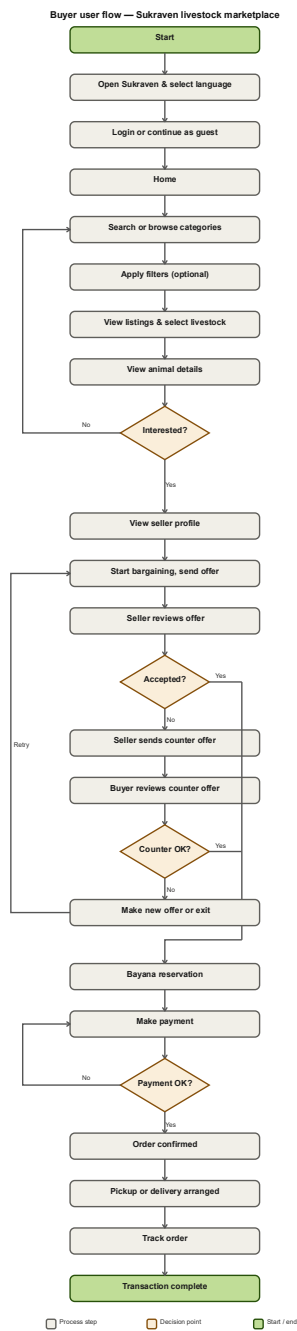


Figure 5.3: Buyer user flow.

The buyer flow (Figure 5.3) traces the path from searching or browsing categories through viewing an animal's details, initiating bargaining, reaching an agreed price, reserving the animal with a Bayana deposit, completing payment, and tracking delivery. A retry loop returns the buyer to browsing if no agreement is reached, and a second retry loop allows the buyer to make a new offer if a seller's counter-offer is not acceptable, directly mirroring the iterative offer/counter-offer pattern observed in the physical Santha (Section 3.1).

5.3.3 Seller Flow

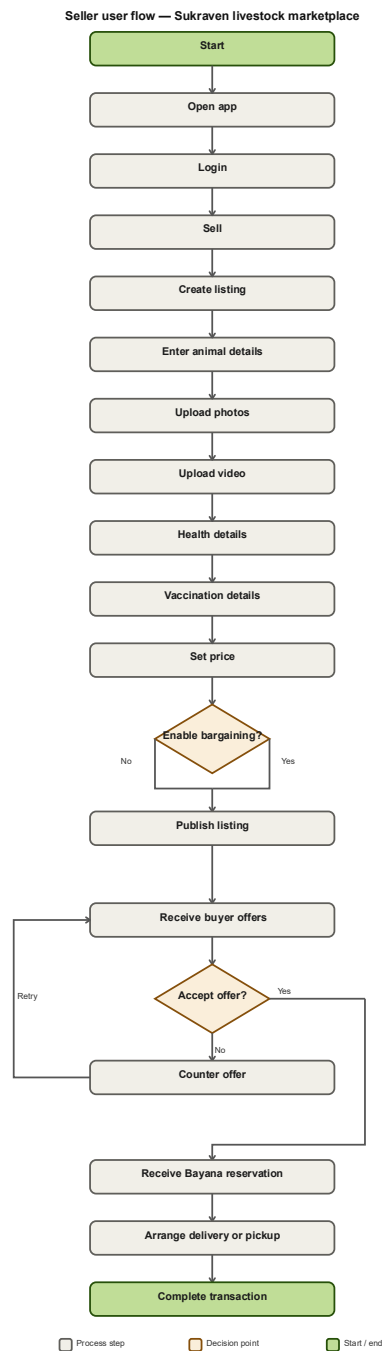


Figure 5.4: Seller user flow.

The seller flow (Figure 5.4) covers the four-step listing-creation process, followed by receiving buyer offers, accepting or countering them, and, on agreement, receiving the Bayana reservation and arranging delivery or pickup. As with the buyer flow, a retry loop allows repeated counter-offers, preserving the back-and-forth negotiation pattern that sellers described as central to achieving a fair price (Section 3.3.3).

5.4 Visual Design System

5.4.1 Logo

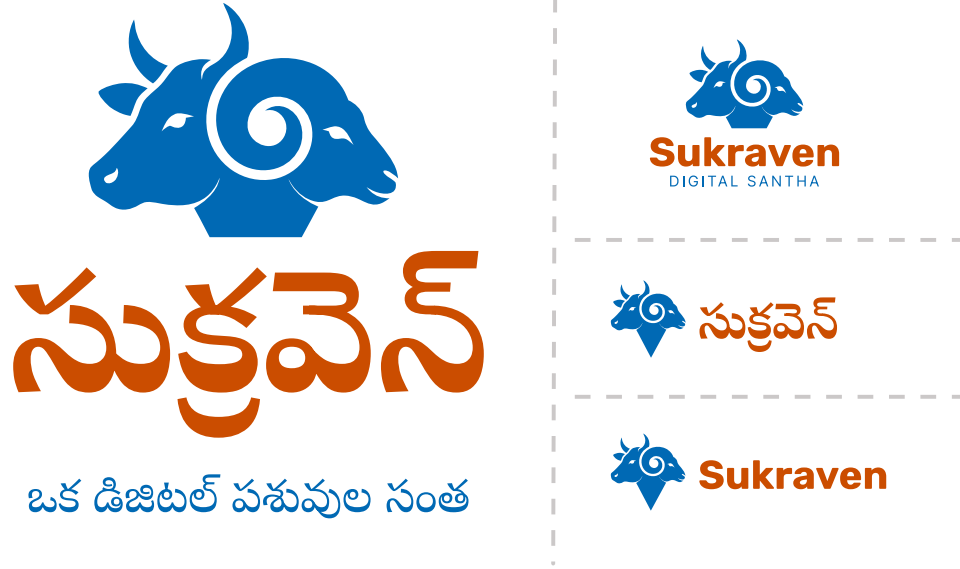


Figure 5.5: Sukraven Logo.

5.4.1 Colour Palette



Figure 5.6: Sukraven colour palette.

The palette (Figure 5.5) pairs a trust-oriented blue (#0A73E0) with a warm, high-visibility orange (#DD5400) drawn from the rural marketplace context, and a distinct green (#00A63E) reserved exclusively for verification and success states, such as the 'vaccinated' and 'deal successful' badges seen throughout the prototype. High-contrast colour coding was chosen deliberately so that verification and status information can be recognised at a glance without relying on text, again reflecting the field-study finding on variable literacy levels (Section 3.2).

5.4.2 Typography

Rubik Bold
Rubik Regular

నోటో సాన్స్ తెలుగు బోల్డ్
(Noto Sans Telugu Bold)

నోటో సాన్స్ తెలుగు రెగ్యులర్
(Noto Sans Telugu Regular)

Figure 5.7: Sukraven typography system.

Sukraven pairs Rubik for English-language text with Noto Sans Telugu for Telugu-script text (Figure 5.6). Noto Sans Telugu was selected specifically for its legibility at small sizes and its comprehensive glyph coverage, which was necessary given that the majority of interface labels in the prototype are rendered in Telugu by default.

5.4.3 Iconography

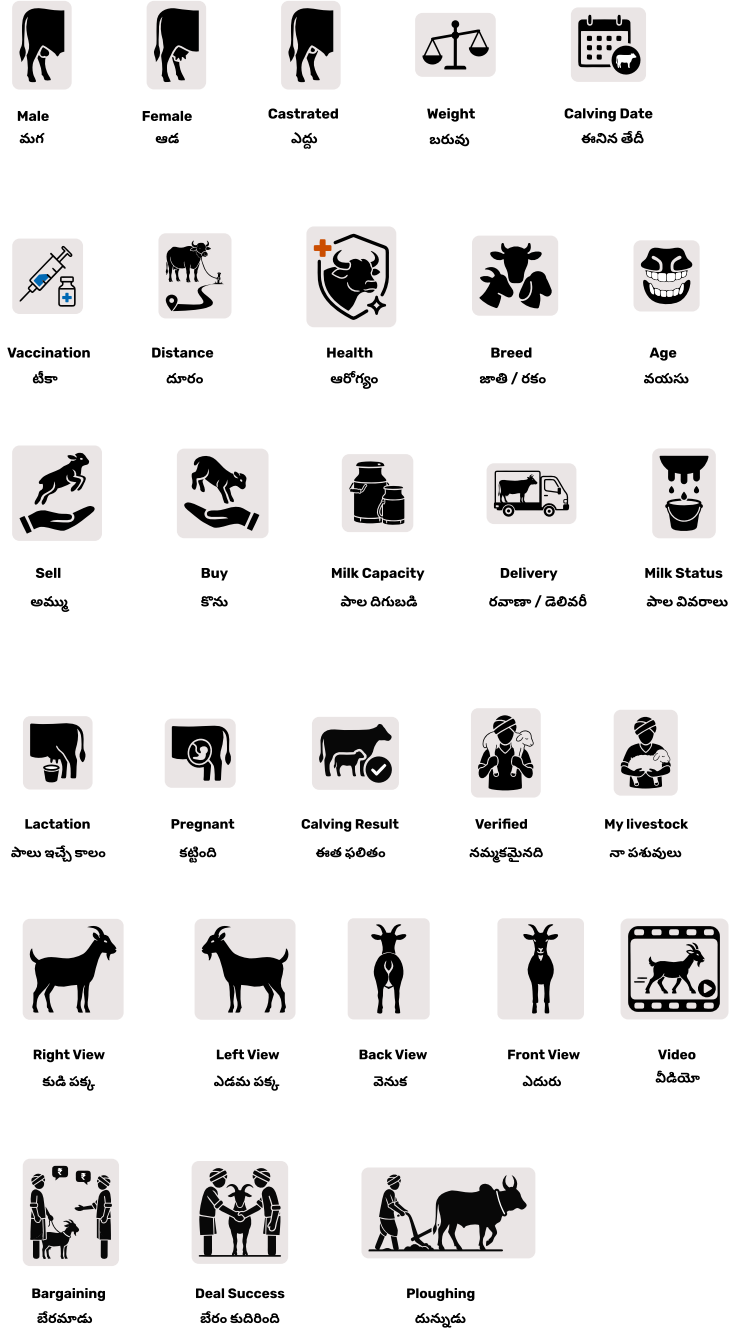


Figure 5.8: Sukraven icon set.

A dedicated icon set (Figure 5.7) was developed to represent recurring animal attributes (male/female/castrated, weight, breed, age, calving date), health and transaction states (vaccination, health, distance, bargaining, deal success), and livestock-specific actions (buy, sell, ploughing capability, milk capacity). Each icon is paired with a short Telugu label throughout the interface, allowing a user to recognise a field's meaning from the icon alone

even where reading the accompanying text is difficult, again directly reflecting Medhi et al.'s (2011) findings on icon-led design for low-literacy users.

5.5 Onboarding and Authentication

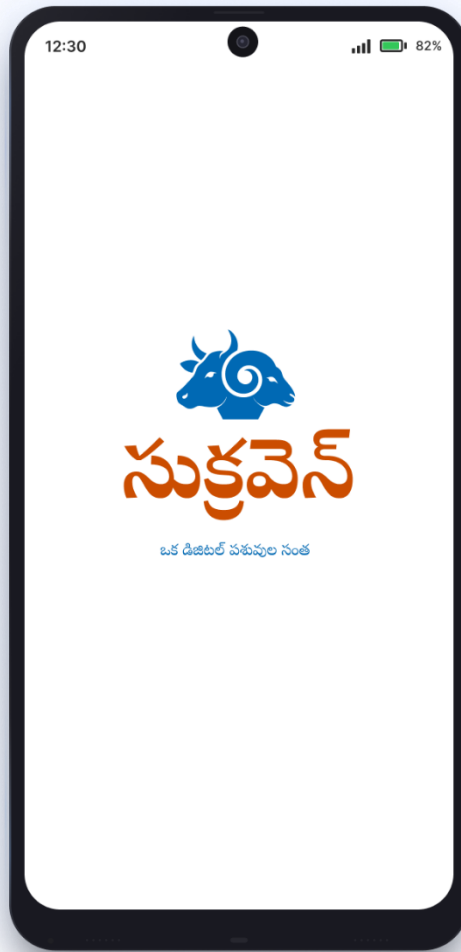


Figure 5.9: Sukraven splash screen.

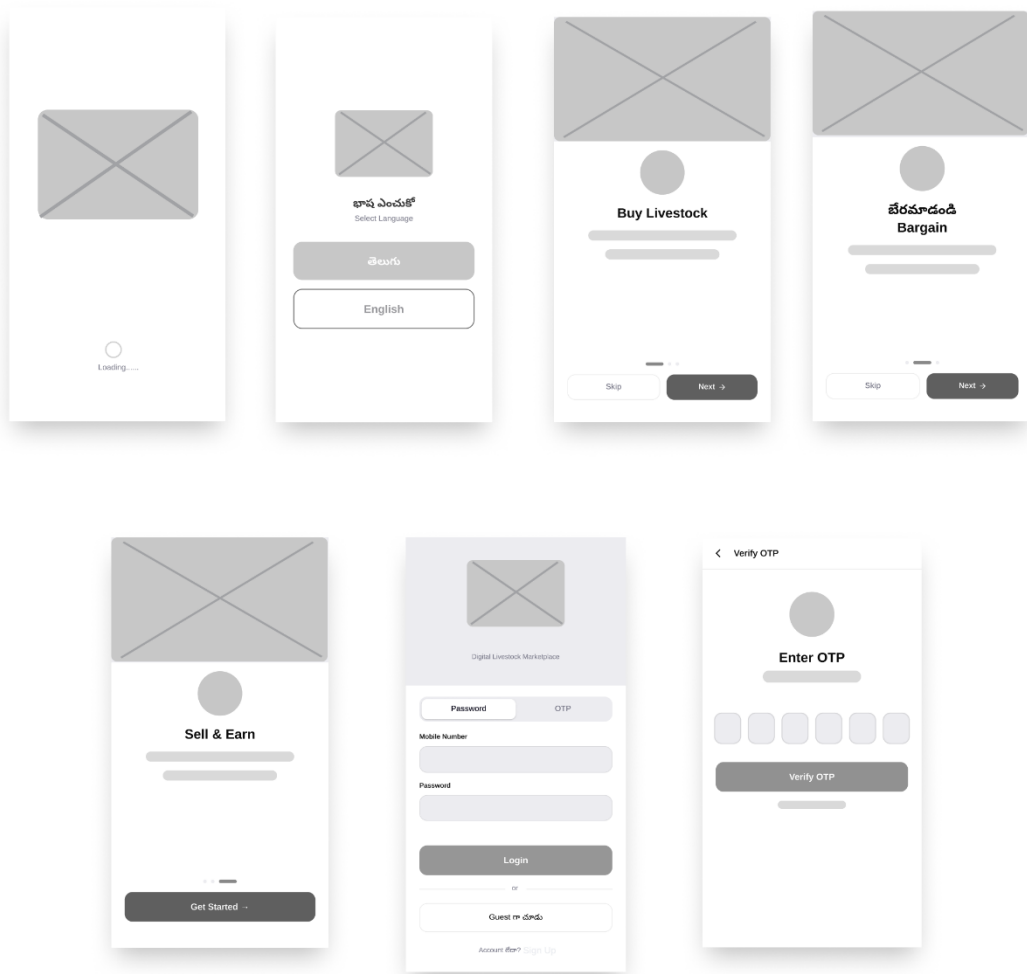


Figure 5.10: Onboarding wireframes.

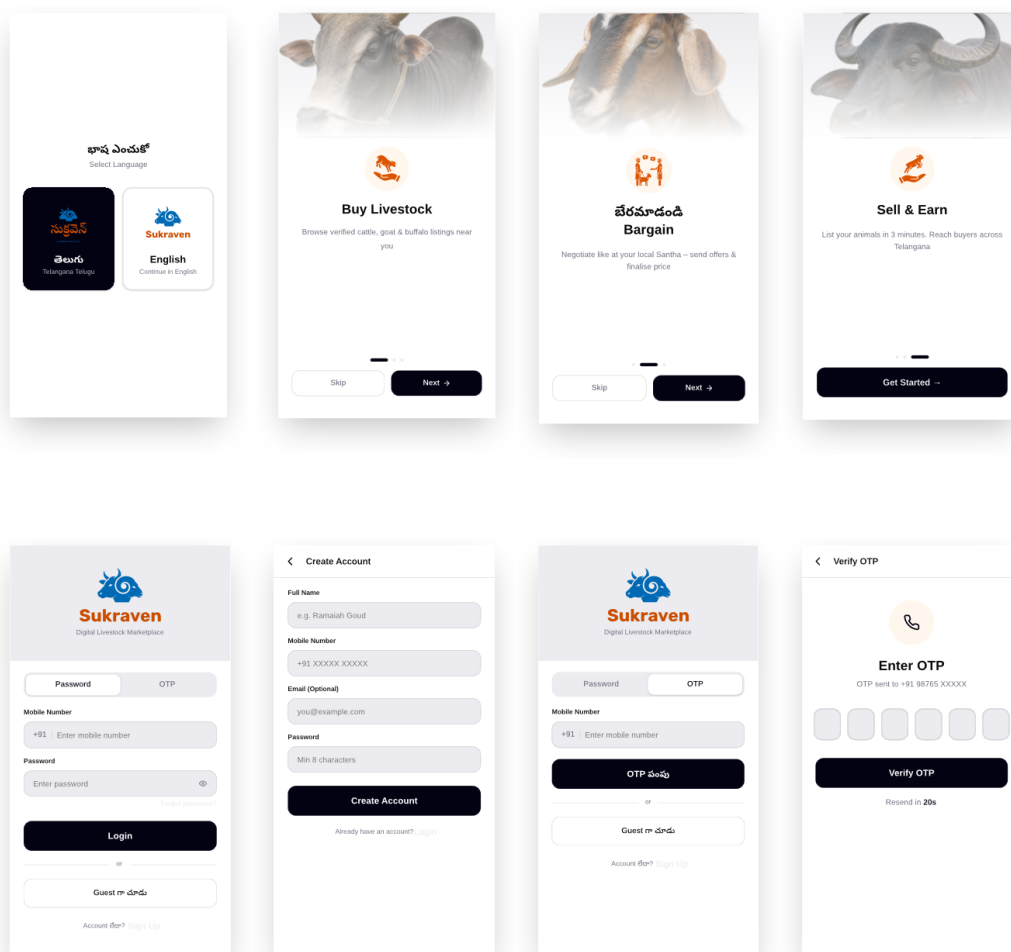


Figure 5.11: Onboarding and authentication screens.

The onboarding sequence (Figures 5.9 and 5.10) opens with a branded splash screen (Figure 5.8) and immediate language selection, followed by a three-screen carousel introducing the application's three core functions -- Buy Livestock, Bargain and Sell & Earn -- each illustrated with a photograph of the relevant animal category rather than descriptive text. Authentication supports both password and OTP-based login, alongside a persistent 'Guest' option, so that a first-time user can explore listings before deciding whether to create an account.

5.6 Home, Search, Browsing and Livestock Detail

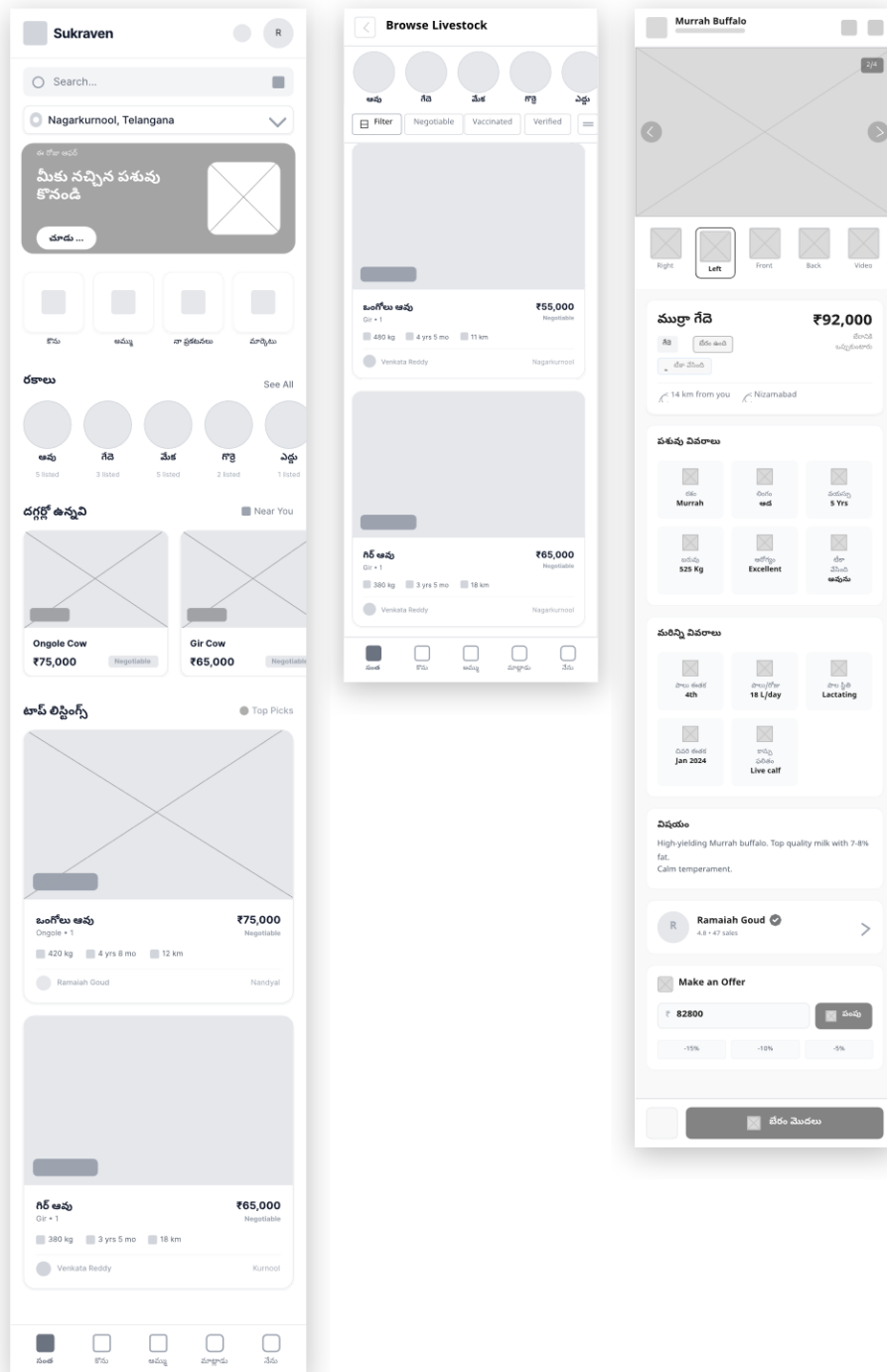


Figure 5.12: Home and livestock-detail wireframes.

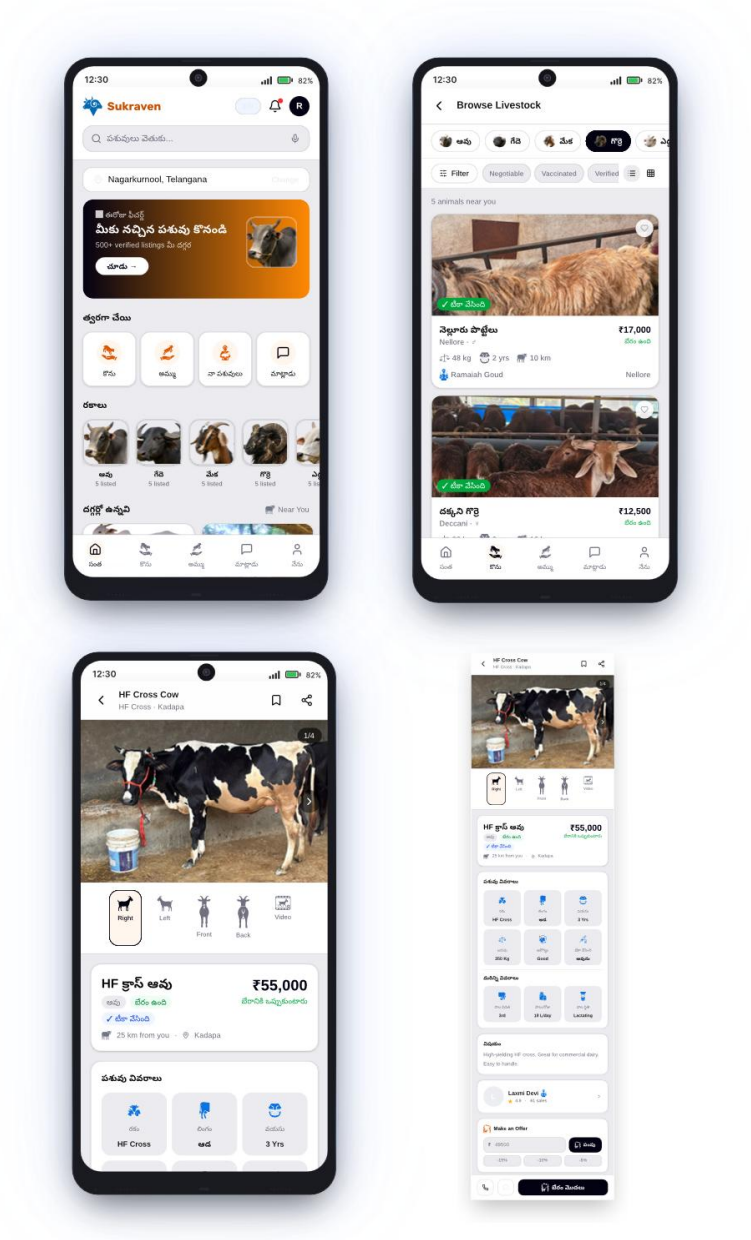


Figure 5.13: Home and livestock-detail screens.

The home screen (Figures 5.11 and 5.12) surfaces category shortcuts (cow, buffalo, goat, sheep, bull), a search bar with voice-input support, a location selector, and 'near you' and 'top listings' carousels, so that a returning user can reach a relevant listing without typing. The livestock detail screen consolidates the information buyers said they needed to evaluate an animal (Section 3.3.2) into a single, icon-led layout: breed, sex, age, weight, health status and vaccination status are shown as labelled icon tiles, four-angle photographs (right, left, front and back) and a video are presented as swipe able tabs, and the seller's rating and sales count are shown directly beneath the listing, immediately above the 'Make an Offer' bargaining entry point.

5.7 Bargaining, Bayana Reservation and Payment

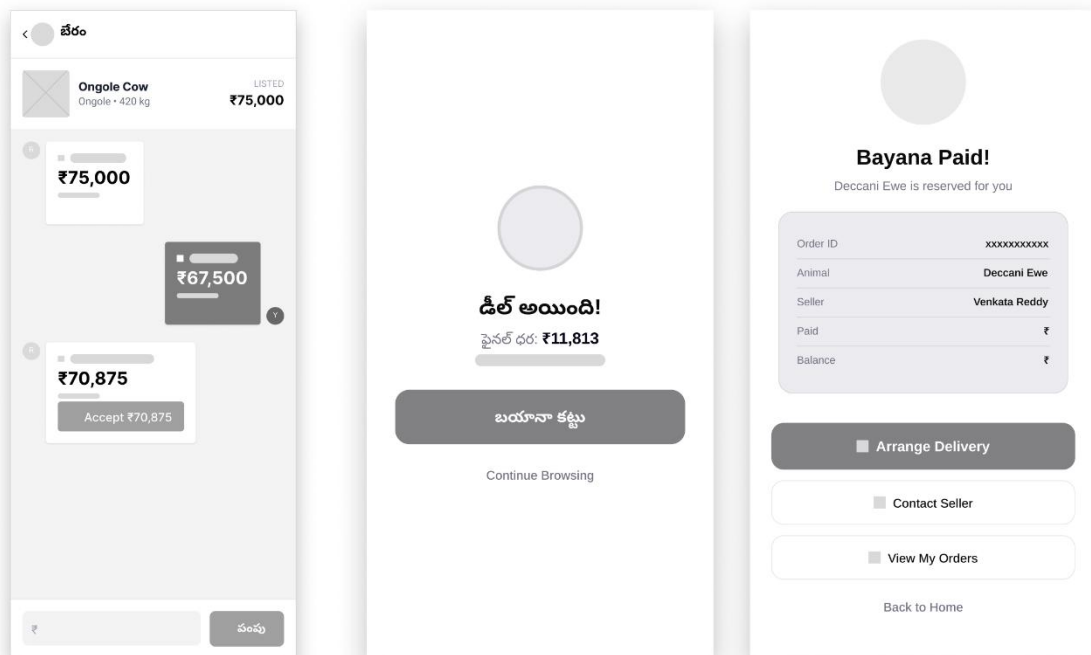


Figure 5.14: Bargaining and Bayana wireframes.

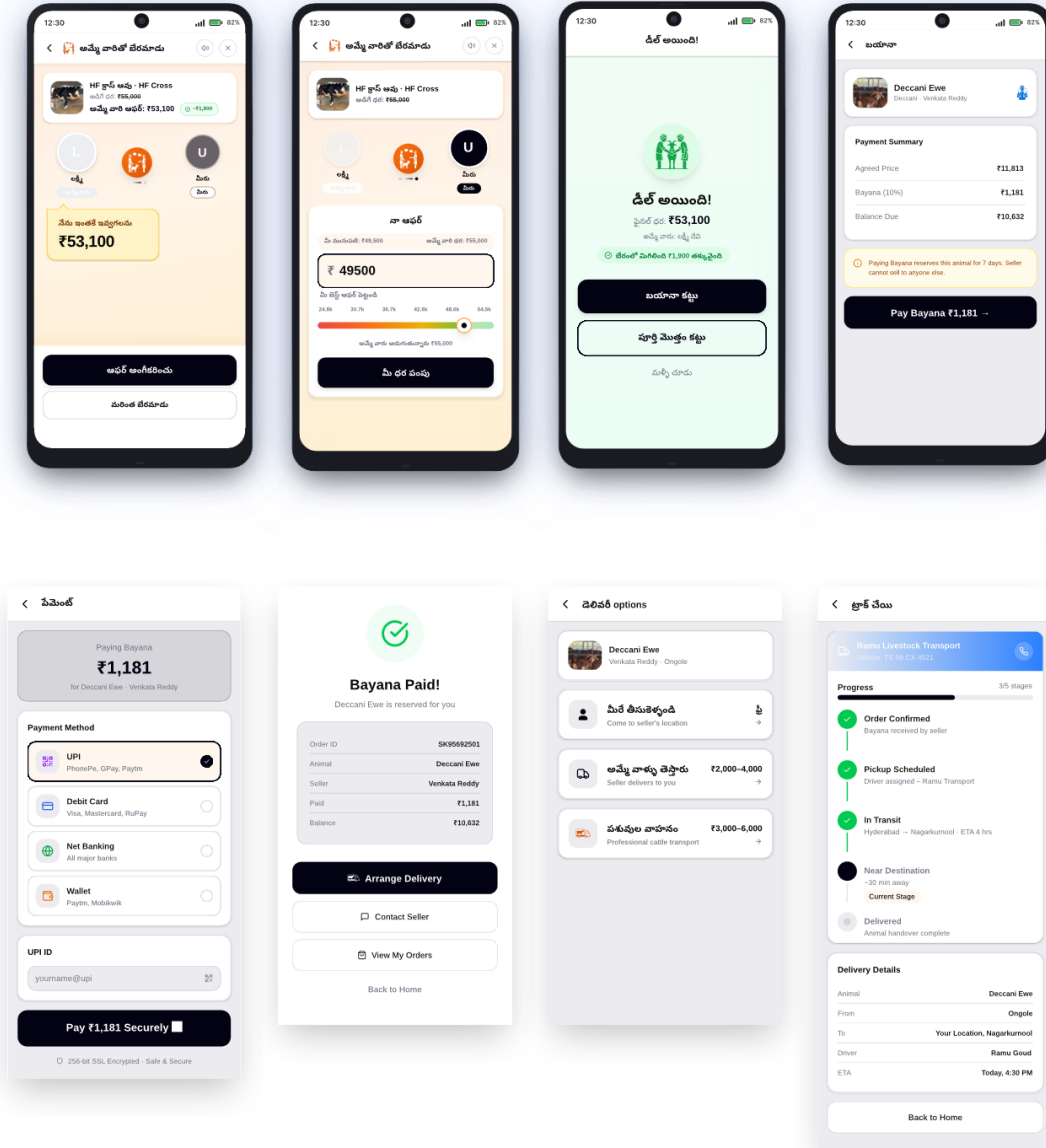


Figure 5.15: Bargaining, payment and delivery screens.

The bargaining screen (Figures 5.13 and 5.14) presents the listed price, the buyer's offer and the seller's counter-offer as a chronological chat-style thread, so that both parties retain a visible record of the negotiation, directly addressing the transparency concerns raised in the field study (Section 3.3.3). On reaching agreement, the buyer is prompted to pay a 10 per cent Bayana (token deposit), which reserves the animal exclusively for seven days; the payment screen supports UPI, debit card, net banking and wallet payment methods, reflecting the field-study finding that UPI was already used by over a third of participants (Section 3.4). Following payment, a delivery-tracking screen presents a five-stage progress indicator (order confirmed, pickup scheduled, in transit, near destination, delivered), addressing the transportation

difficulties that both buyers and sellers identified as a recurring pain point in their journey maps (Section 3.6).

5.8 Listing Creation and Seller Dashboard

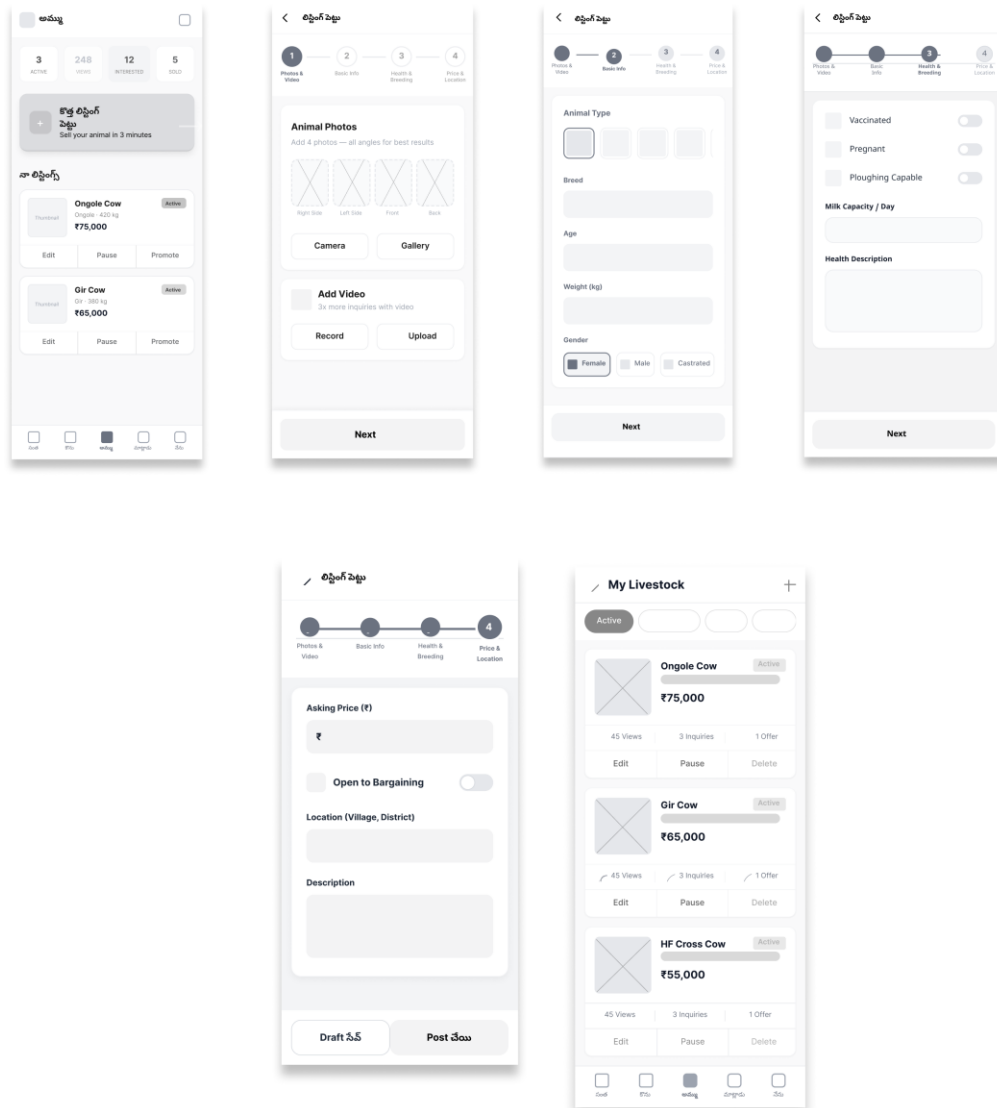


Figure 5.16: Listing-creation wireframes.

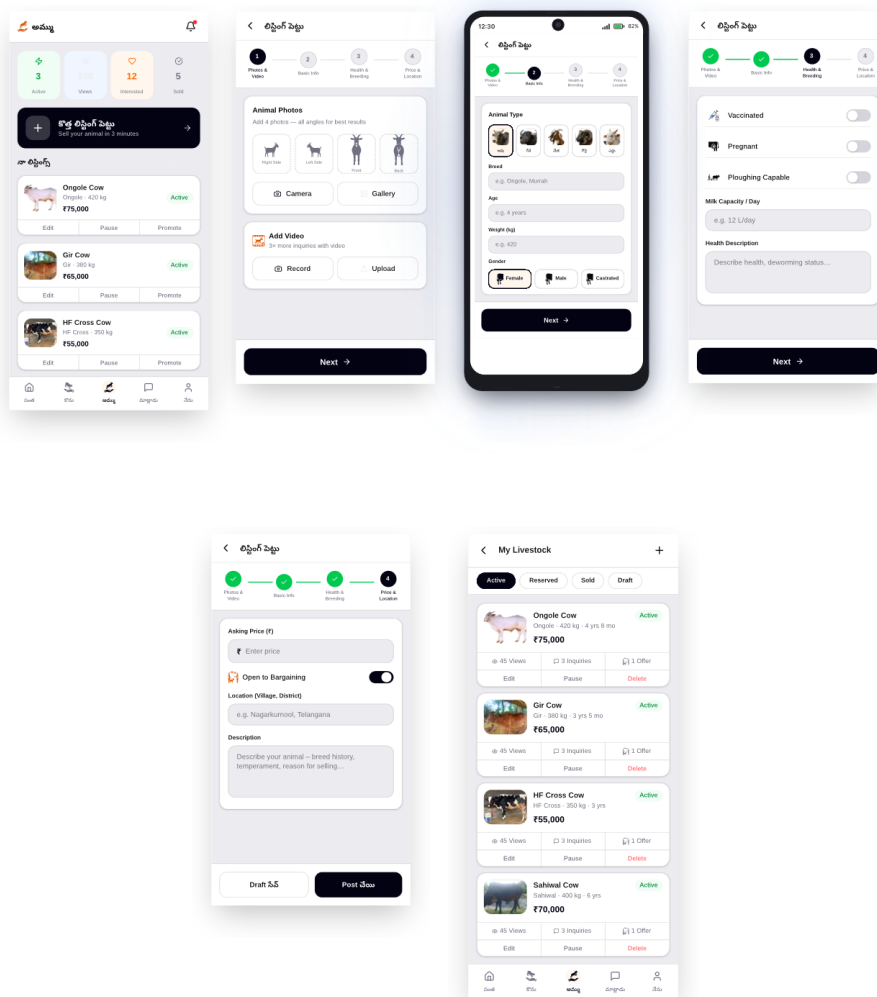


Figure 5.17: Listing-creation and dashboard screens.

The listing-creation flow (Figures 5.15 and 5.16) is organised into four steps, tracked by a persistent progress indicator: photos and video (compulsory right, left, front and back angles); basic information (animal type, breed, age, weight, gender); health and breeding details (vaccination, pregnancy, ploughing capability, milk capacity); and price and location, including a toggle to enable bargaining on the listing. Requiring four-angle photography before a listing can be published operationalises the trust-building requirement identified in Section 3.4, where livestock photographs and videos were among the most frequently requested trust features. The My Livestock dashboard (Figure 5.16, bottom right) lets a seller review view, inquiry and offer counts for each active listing and manage listings across active, reserved, sold and draft states, directly addressing sellers' reported difficulty finding genuine buyers (Section 3.3.1) by surfacing engagement metrics for each listing.

5.9 Search, Chat and Profile Management

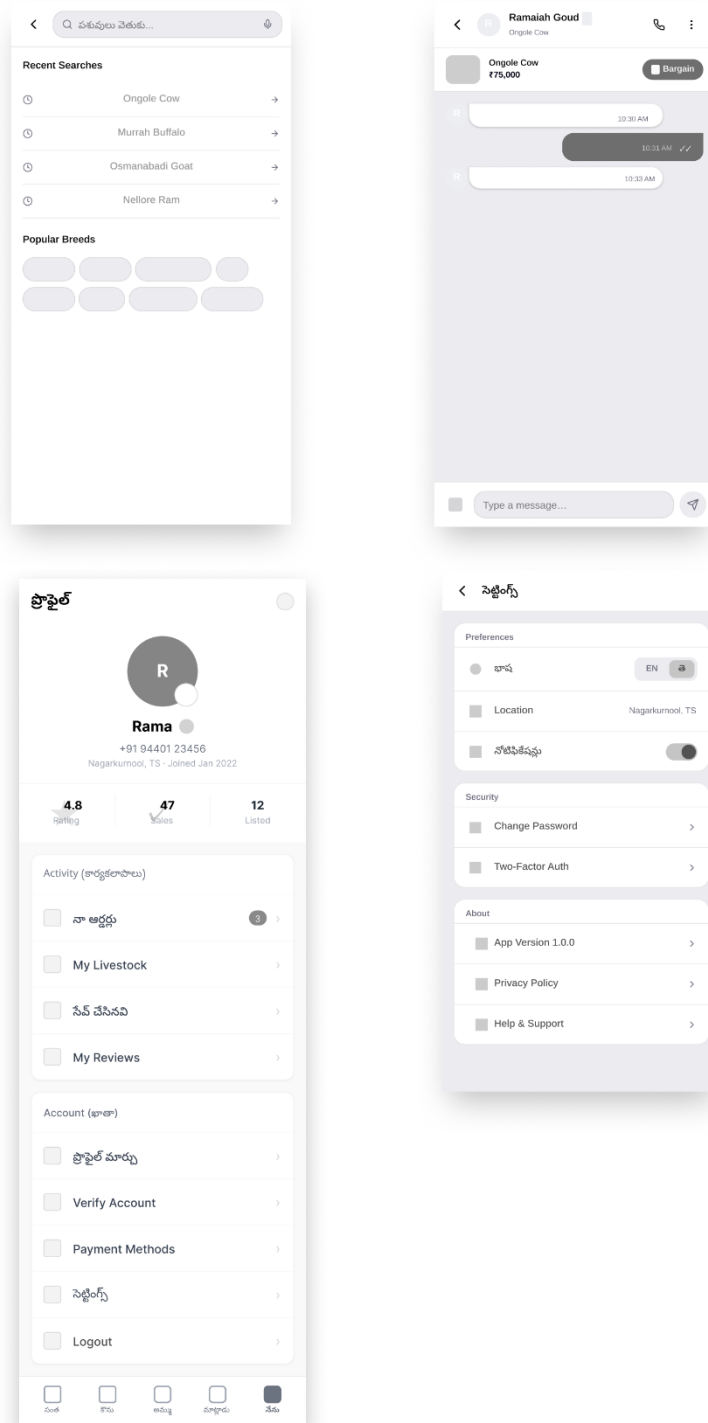


Figure 5.18: Search, chat and profile wireframes.

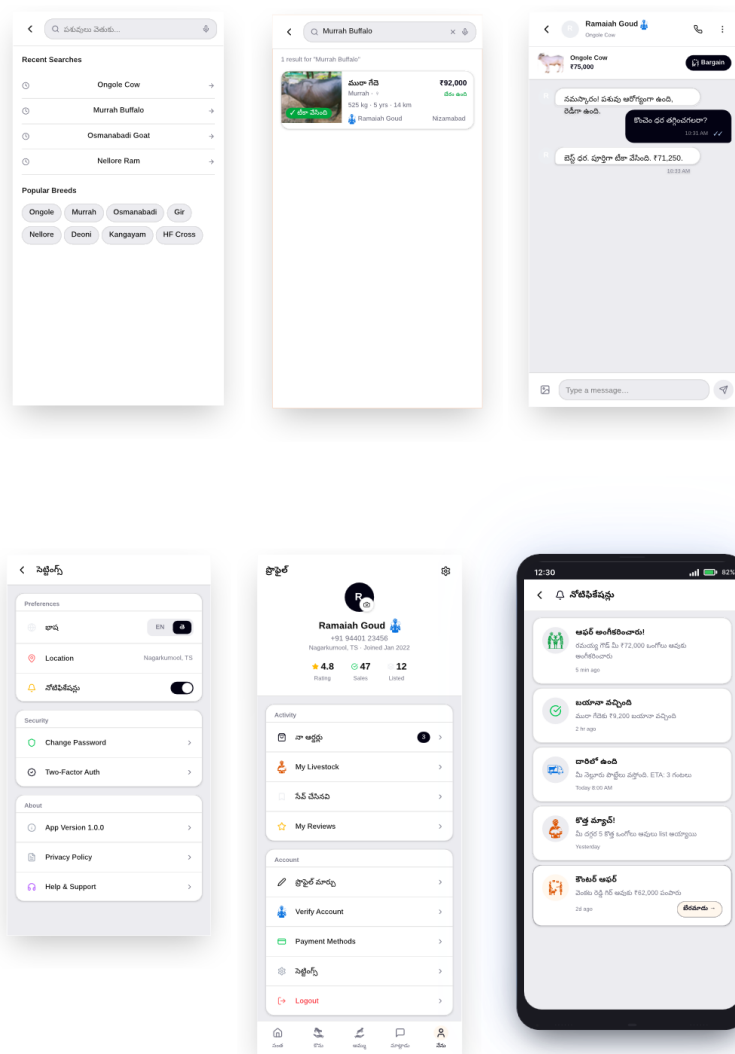


Figure 5.19: Search, chat and profile screens.

The search screen (Figures 5.17 and 5.18) surfaces recent searches and popular breed shortcuts (Ongole, Murrah, Osmanabadi, Gir, Nellore) so that a user can find a specific breed without typing a full query, addressing the difficulty finding suitable livestock reported by buyers (Section 3.3.2). The chat screen anchors every conversation to the specific listing under negotiation, with a persistent 'Bargain' action visible at the top of the thread. The profile and settings screens expose language, location and notification preferences, along with account verification, saved listings, orders and review history, consolidating the account-management functions that both buyers and sellers need to track an ongoing or completed transaction.

Chapter 6 Conclusion, Limitations and Future Scope

6.1 Conclusion

This thesis set out to design and evaluate an inclusive digital marketplace for livestock trading in Southern Telangana that preserves bargaining while improving accessibility, transparency and trust. A field study of 82 buyers and sellers across the Nagarkurnool, Wanaparthy and Mahabubnagar districts established that bargaining is universal and highly valued (100% of participants trade without fixed pricing; 76.8% rate bargaining as very important), that trust between trading parties rests on familiarity and direct verification rather than formal credentials, and that while smartphone ownership is reasonably high (69.5%), prior experience with any digital trading platform is low (24.4%). These findings directly shaped the design of Sukraven, a bilingual Telugu/English mobile application that digitises the Santha's negotiation-centred structure through an explicit offer/counter-offer bargaining flow, verified profiles, compulsory four-angle livestock photography and video, seller ratings and sales history, and a Bayana token-deposit reservation mechanism.

Evaluated with 18 independent buyers and sellers, the resulting high-fidelity prototype achieved a mean System Usability Scale score of 80.42 out of 100 -- a 'Good' rating on standard usability benchmarks (Bangor et al., 2008), reported subject to the reverse-worded-item validity caveat discussed in Section 3.7.1 and Section 6.2 -- alongside a mean trust-perception score of 4.31 out of 5 and a mean user-satisfaction score of 4.29 out of 5. Read together with the qualitative field-study findings, these results support the central proposition of this thesis: that a digital marketplace for rural livestock trading is more likely to be adopted and trusted when it digitises, rather than replaces, the negotiation and familiarity-based trust practices that already define how buyers and sellers in this region trade, and when its interface is designed from the outset around vernacular language and low-literacy-friendly, icon-led interaction patterns.

More broadly, this work contributes an empirically grounded design case for inclusive digital marketplace design in an agricultural, rural Indian context, complementing existing literature on low-literacy mobile interface design (Medhi et al., 2011) and on the adoption barriers facing national digital agricultural platforms such as e-NAM (Manogna, 2025), by showing how field-derived requirements around bargaining and trust can be translated into specific, evaluable interface decisions for the previously under-served case of livestock trading.

6.2 Limitations

Several limitations should be considered when interpreting the findings of this thesis.

- Sample size of the usability evaluation: the quantitative usability, trust and satisfaction results are based on 18 participants (10 buyers, 8 sellers). While sufficient to identify clear directional patterns and consistent with common practice for formative usability evaluation, this sample is too small to support robust statistical inference or to generalise with confidence to the wider population of livestock traders in Southern Telangana.
- Demographic profile of the usability-evaluation participants: the 18 usability-evaluation participants were recruited using the same purposive-sampling approach, inclusion criteria and study districts as the field study (Section 2.3), but their individual demographic characteristics (age band, education level, phone ownership) were not separately tabulated in this thesis in the same form as Table 3.1 for the 82 field-study participants. This information was not recorded on a structured instrument during the May 2026 sessions and could not be reconstructed from the researcher's surviving notes for this thesis; it is therefore reported here as a genuine and currently unresolved gap in the evidence base, not a routine item still to be completed. Because the thesis's central claim concerns usability for low-literacy users specifically, the reader cannot verify from the data reported here whether the 80.42 SUS score was earned with a sample that included participants with no formal education, or with a more literate convenience subset of the same purposive-sampling pool; the corresponding claim in Section 4.3 is qualified accordingly, and closing this gap (for example by contacting participants to record these details retrospectively, where feasible and with fresh consent) should be treated as a priority before the thesis defence.
- Response pattern on reverse-worded SUS items: across the 18 usability-evaluation participants, no response to any of the five negatively-worded SUS items (SUS2, 4, 6, 8, 10) fell in the 'Agree' or 'Strongly Agree' categories. The instrument was administered orally in Telugu in the same manner as the field-study questionnaire, and these items, like all others, were read to participants exactly as translated, without rewording, facilitator restatement or a visual response aid (Section 2.5). The absence of any lapse in polarity-tracking across 90 responses from 18 participants is therefore reported as an observed feature of this dataset notable, in that it exceeds what is typically expected when the SUS is administered orally in comparable rural, low-literacy settings. Whether this reflects genuine comprehension of the reversed items or an artefact of oral delivery and translation cannot be determined conclusively from the data available for this thesis, and this thesis does not assert either explanation over the other. The full item-level response matrix for these five items is reproduced and discussed in Section 3.7.1 (Table 3.6a): responses vary across Strongly Disagree, Disagree and Neutral without ever reaching Agree or Strongly Agree, which is more consistent with differentiated engagement than with a uniform

response strategy, but this falls short of independent confirmation. Session recordings were not made and no repeat administration was carried out; either would be needed to resolve the question conclusively, and until then the SUS score of 80.42 reported throughout this thesis (Abstract, Section 3.7.1, Section 4.3, Section 6.1) should be read together with this caveat rather than as an unqualified figure.

- Retrospective and, at the time of writing, unresolved ethical clearance: consent for both the December 2025 field study and the May 2026 usability evaluation was obtained orally rather than through signed consent forms, and formal institutional ethics clearance was not obtained prior to data collection. The oral consent actually sought was documented, not merely verbal: the script covering purpose, voluntariness, the right to withdraw without penalty, and data use and storage is reproduced in Appendix C. An Ethical Clearance Form documenting the study in full was submitted to the PDPM IIITDM Jabalpur Institutional Ethics Committee on 29 July 2026 for retrospective review, but as of the submission of this thesis the Committee had not yet returned a decision, and no approval reference number had been assigned. This is therefore reported here as an open matter rather than a resolved limitation: the ethical status of the complete dataset (82 field-study and 18 usability-evaluation participants) depends on the outcome of that review, and obtaining and citing that outcome before the thesis defence is treated as a critical priority rather than an optional improvement.
- Prototype evaluation, not field deployment: the usability evaluation was conducted on a high-fidelity interactive prototype in a moderated setting rather than through live, unmoderated use of a deployed application handling real transactions, real payments and real livestock deliveries over time. Findings on satisfaction and trust may therefore differ once the application is used for actual, financially consequential transactions.
- Regional and linguistic scope: both study phases drew participants exclusively from the Nagarkurnool, Wanaparthy and Mahabubnagar districts of Southern Telangana, trading predominantly in Telangana Telugu. The findings and design decisions reported here may not transfer directly to livestock-trading contexts in other regions, languages or market structures.
- Absence of a comparison condition: the usability evaluation assessed the Sukraven prototype in isolation rather than in direct, controlled comparison with an existing livestock-trading application or with the physical Santha experience itself, limiting the strength of any claim that Sukraven is more usable or trustworthy than available alternatives.
- Self-reported measures: SUS, trust-perception and satisfaction scores are all self-reported and were collected immediately after a single moderated session; they do not capture longer-term behavioural outcomes such as sustained use, actual transaction completion rates, or real financial risk-taking.

6.3 Future Scope

Building on this thesis, future work could extend the usability evaluation to a larger and more diverse sample of buyers and sellers, including participants from additional districts and livestock categories not covered here, to test the generalisability of the current findings. A field deployment of Sukraven handling genuine listings, bargaining and Bayana payments would allow satisfaction, trust and usability to be assessed under real transactional stakes rather than in a moderated prototype session, and would allow the seller-side listing-creation flow -- identified in Section 4.4 as the likely source of the buyer-seller usability gap -- to be iterated further. Longitudinal research tracking the same buyers and sellers across multiple market cycles could also clarify whether trust in the platform builds over repeated use, as it does in the physical Santha, and whether the improvements in transparency and price information reported here translate into measurably fairer prices over time.

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Appendix A: Field-Study Questionnaire (English)

The following questionnaire was administered orally in Telugu (see Appendix B) to all 82 field-study participants in December 2025; the English version below was used for translation, data recording and analysis.

Section A: Participant Profile

1. Role

- ☐ Buyer
- ☐ Seller
- ☐ Both Buyer & Seller

2. Gender

- ☐ Male
- ☐ Female
- ☐ Prefer not to say

3. Age Group

- ☐ Below 25 years
- ☐ 25–40 years
- ☐ 41–60 years
- ☐ Above 60 years

4. Education Level

- ☐ No formal education
- ☐ Primary
- ☐ Secondary
- ☐ Higher Secondary
- ☐ Graduate and above

5. Primary Livestock Traded

- ☐ Sheep
- ☐ Goat
- ☐ Cow
- ☐ Buffalo
- ☐ Mixed

6. Years of Experience in Livestock Trading

_____ Years

7. Frequency of Visiting Weekly Santha

- ☐ Every week
- ☐ Twice a month
- ☐ Occasionally

Section B: Livestock Market Experience

For Sellers

8. How do you determine the selling price of your livestock?

9. What are the major challenges you face while selling livestock? (Select all that applies.)

- ☐ Receiving lower price than expected
- ☐ Difficulty finding genuine buyers
- ☐ Transportation difficulties
- ☐ Delay in selling livestock
- ☐ Lack of information about market prices
- ☐ Animal health concerns
- ☐ Other _____

10. Are you satisfied with the final selling price?

- ☐ Yes
- ☐ No

If No, why?

11. What expenses do you usually incur while selling livestock? (Select all that applies.)

- ☐ Transportation
- ☐ Market entry fee
- ☐ Animal feeding
- ☐ Veterinary expenses
- ☐ Other _____

12. Have you ever experienced unfair practices during livestock trading?

- ☐ Yes
- ☐ No

If yes, explain.

For Buyers

13. How do you evaluate livestock before purchasing?

14. What are the major challenges you face while buying livestock? (Select all that applies)

- ☐ Unclear pricing
- ☐ Difficulty assessing livestock quality
- ☐ Lack of livestock information
- ☐ Difficulty finding suitable livestock
- ☐ Transportation
- ☐ Trust concerns
- ☐ Other _____

15. Are you confident about the quality of livestock before purchasing?

- ☐ Yes
- ☐ No

16. How is the price usually finalized?

- ☐ Through direct bargaining
- ☐ Fixed price

- Mutual discussion

Common Questions

17. Is there any fixed pricing system in your local livestock market?

- Yes
- No

18. How important is bargaining during livestock transactions?

- Very Important
- Moderately Important
- Slightly Important
- Not Important

19. What factors help you trust the other person during livestock trading?

20. What improvements would you like to see in the current livestock trading system?

Section C: Technology Awareness & Digital Readiness

21. Do you own a mobile phone?

- Smartphone
- Basic phone
- No

22. How often do you use your mobile phone?

- Daily
- Occasionally
- Rarely

23. What do you mainly use your mobile phone for? (Select all that apply.)

- Calls
- WhatsApp
- YouTube
- UPI Payments
- Social Media
- Other _____

24. Have you ever used an online platform to buy or sell livestock?

- Yes
- No

25. Would you be willing to use a mobile application for livestock trading?

- Yes
- Maybe
- No

26. What difficulties might you face while using such an application? (Select all that apply.)

- Language
- Reading difficulty

- Internet connectivity
- Lack of smartphone knowledge
- Trust issues
- Other _____

27. Which features would increase your trust in a livestock trading application? (Select all that apply.)

- Verified user profiles
- Multiple livestock photographs
- Livestock videos
- Health & vaccination details
- Transparent pricing
- User ratings/reviews
- Other _____

28. Which features would you expect in such an application? (Select all that apply.)

- Digital bargaining
- Livestock details (Breed, Age, Weight, Health)
- Search and Filters
- Distance to seller
- Verified profiles
- Favourite listings
- Price guidance
- Other _____

Section D: Perception Statements

Please indicate your level of agreement with each statement.

- 1 = Strongly Disagree
- 2 = Disagree
- 3 = Neutral
- 4 = Agree
- 5 = Strongly Agree

29. Finding genuine buyers or sellers in livestock markets is difficult.

30. Bargaining is essential for obtaining a fair price in livestock trading.

31. It is difficult to assess livestock quality before purchase.

32. I am satisfied with the current livestock trading process.

33. I feel confident using smartphone applications.

34. A digital livestock marketplace would improve livestock trading.

35. Verified user profiles would increase my trust in a livestock trading application.

36. Detailed livestock information (breed, age, weight and health) would help me make better buying or selling decisions.

Appendix B: Field-Study Questionnaire (Telugu)

This is the Telugu-language form of the questionnaire in Appendix A, as read aloud to participants in the field.

విభాగం A: పాల్గొనేవారి వివరాలు

1. పాత్ర

- కొనుగోలుదారు
- విక్రేత (అమ్మకందారు)
- కొనుగోలుదారు మరియు విక్రేత (రెండు)

2. లింగం

- పురుషుడు
- స్త్రీ
- చెప్పడానికి ఇష్టపడను

3. వయస్సు

- 25 సంవత్సరాల కంటే తక్కువ
- 25–40 సంవత్సరాలు
- 41–60 సంవత్సరాలు
- 60 సంవత్సరాల కంటే ఎక్కువ

4. విద్యాభ్యాసం

- చదువుకోలేదు
- ప్రాథమిక విద్య
- మాధ్యమిక విద్య
- ఉన్నత మాధ్యమిక విద్య
- డిగ్రీ మరియు అంతకంటే ఎక్కువ

5. ప్రధానంగా వ్యాపారం చేసే పశువులు

- గొర్రెలు
- మేకలు
- ఆవులు
- గేదెలు
- మిశ్రమం (అన్నీ)

6. పశువుల వ్యాపారంలో అనుభవం (సంవత్సరాలలో)

_____ సంవత్సరాలు

7. వారపు సంతకు ఎంత తరచుగా వెళ్తారు?

- ప్రతి వారం
- నెలకు రెండుసార్లు
- అప్పుడప్పుడు

విభాగం B: పశువుల మార్కెట్ అనుభవం
విక్రేతలకు (అమ్మకందారులకు)

8. మీ పశువుల అమ్మకపు ధరను ఎలా నిర్ణయిస్తారు?

9. పశువులను విక్రయించేటప్పుడు మీరు ఎదుర్కొనే ప్రధాన సవాళ్లు ఏమిటి? (వర్తించేవన్నీ ఎంచుకోండి.)

- ఆశించిన దానికంటే తక్కువ ధర రావడం
- సరైన కొనుగోలుదారులను కనుగొనడం కష్టం కావడం
- రవాణా ఇబ్బందులు
- పశువుల అమ్మకంలో జాప్యం (ఆలస్యం)
- మార్కెట్ ధరల గురించి సమాచారం లేకపోవడం
- పశువుల ఆరోగ్య సమస్యలు
- ఇతరాలు _____

10. తుది అమ్మకపు ధరతో మీరు సంతృప్తి చెందారా?

- అవును
- కాదు

కాకపోతే, ఎందుకు?

11. పశువులను విక్రయించేటప్పుడు మీకు సాధారణంగా ఏయే ఖర్చులు అవుతాయి? (వర్తించేవన్నీ ఎంచుకోండి.)

- రవాణా
- మార్కెట్ ప్రవేశ రుసుము (ఎంట్రీ ఫీజు)
- పశువుల మేత
- వెటర్నరీ (వైద్య) ఖర్చులు
- ఇతరాలు _____

12. పశువుల వ్యాపారంలో మీరు ఎప్పుడైనా మోసాలకు గురయ్యారా (అన్యాయమైన పద్ధతులను ఎదుర్కొన్నారా)?

- అవును
- లేదు

అవును అయితే, వివరించండి.

కొనుగోలుదారులకు

13. పశువులను కొనుగోలు చేసే ముందు వాటిని ఎలా అంచనా వేస్తారు?

14. పశువులను కొనుగోలు చేసేటప్పుడు మీరు ఎదుర్కొనే ప్రధాన సవాళ్లు ఏమిటి? (వర్తించేవన్నీ ఎంచుకోండి)

- ధరలో స్పష్టత లేకపోవడం
- పశువుల నాణ్యతను అంచనా వేయడం కష్టం కావడం
- పశువుల గురించి తగిన సమాచారం లేకపోవడం
- మీకు అనువైన పశువులను కనుగొనడం కష్టం కావడం
- రవాణా
- నమ్మకానికి సంబంధించిన సమస్యలు

- ఇతరాలు _____

15. కొనుగోలు చేయడానికి ముందు పశువుల నాణ్యత పట్ల మీకు నమ్మకం ఉంటుందా?

- అవును
- లేదు

16. సాధారణంగా ధర ఎలా ఖరారు అవుతుంది?

- ప్రత్యక్ష బేరసారాల ద్వారా
- నిర్ణీత (ఫిక్స్డ్) ధర
- పరస్పర చర్చ ద్వారా

సాధారణ ప్రశ్నలు

17. మీ స్థానిక పశువుల మార్కెట్‌లో ఏదైనా నిర్ణీత (ఫిక్స్డ్) ధరల విధానం ఉందా?

- అవును
- లేదు

18. పశువుల లావాదేవీలలో బేరసారాలు ఎంత ముఖ్యమైనవి?

- చాలా ముఖ్యం
- కొంతమేరకు ముఖ్యం
- కొద్దిగా ముఖ్యం
- ముఖ్యం కాదు

19. పశువుల వ్యాపారంలో ఎదుటి వ్యక్తిని నమ్మడానికి ఏ అంశాలు మీకు సహాయపడతాయి?

20. ప్రస్తుత పశువుల వ్యాపార విధానంలో మీరు ఎలాంటి మార్పులు/మెరుగుదలలు చూడాలనుకుంటున్నారు?

విభాగం C: సాంకేతిక అవగాహన & డిజిటల్ సన్నద్ధత

21. మీ వద్ద మొబైల్ ఫోన్ ఉందా?

- స్మార్ట్ ఫోన్
- బేసిక్ (సాధారణ) ఫోన్
- లేదు

22. మీరు మీ మొబైల్ ఫోన్‌ను ఎంత తరచుగా ఉపయోగిస్తారు?

- ప్రతిరోజూ
- అప్పుడప్పుడు
- అరుదుగా

23. మీరు మీ మొబైల్ ఫోన్‌ను ముఖ్యంగా దేని కోసం ఉపయోగిస్తారు? (వర్తించేవన్నీ ఎంచుకోండి.)

- కాల్స్
- వాట్సాప్ (WhatsApp)
- యూట్యూబ్ (YouTube)
- UPI చెల్లింపులు (PhonePe/GPay వంటివి)

- సోషల్ మీడియా
- ఇతరాలు _____

24. పశువులను కొనడానికి లేదా అమ్మడానికి మీరు ఎప్పుడైనా ఆన్‌లైన్ ప్లాట్‌ఫారమ్‌ను ఉపయోగించారా?

- అవును
- లేదు

25. పశువుల వ్యాపారం కోసం మొబైల్ అప్లికేషన్ (యాప్)ను ఉపయోగించడానికి మీరు ఆసక్తి చూపుతారా?

- అవును
- బహుశా
- లేదు

26. అలాంటి అప్లికేషన్‌ను ఉపయోగించేటప్పుడు మీరు ఎలాంటి ఇబ్బందులు ఎదుర్కోవచ్చు? (వర్తించేవన్నీ ఎంచుకోండి.)

- భాషా సమస్య
- చదవడంలో ఇబ్బంది
- ఇంటర్నెట్ కనెక్టివిటీ సమస్య
- స్మార్ట్‌ఫోన్ పరిజ్ఞానం లేకపోవడం
- నమ్మకం లేకపోవడం (ట్రస్ట్ ఇష్యూస్)
- ఇతరాలు _____

27. పశువుల వ్యాపార అప్లికేషన్‌లో ఏ ఫీచర్లు ఉంటే మీ నమ్మకం పెరుగుతుంది? (వర్తించేవన్నీ ఎంచుకోండి.)

- వెరిఫైడ్ యూజర్ ప్రొఫైల్స్ (ధృవీకరించబడిన ఖాతాలు)
- పశువులకు సంబంధించిన అనేక ఫోటోలు
- పశువుల వీడియోలు
- ఆరోగ్యం & వ్యాక్సినేషన్ (టీకాల) వివరాలు
- పారదర్శకమైన ధరలు
- యూజర్ రేటింగ్స్/రివ్యూలు
- ఇతరాలు _____

28. అలాంటి అప్లికేషన్‌లో మీరు ఎలాంటి ఫీచర్లను ఆశిస్తారు? (వర్తించేవన్నీ ఎంచుకోండి.)

- డిజిటల్ బేరసారాలు
- పశువుల వివరాలు (జాతి, వయస్సు, బరువు, ఆరోగ్యం)
- సెర్చ్ మరియు ఫిల్టర్లు
- అమ్మకందారుకి ఉన్న దూరం
- వెరిఫైడ్ ప్రొఫైల్స్
- ఇష్టమైన జాబితాలు (ఫేవరైట్స్)
- ధరల మార్గదర్శకత్వం
- ఇతరాలు _____

విభాగం D: అభిప్రాయ ప్రకటనలు

దయచేసి ప్రతి వాక్యంతో మీ అంగీకార ఫోయిని సూచించండి.

- 1 = పూర్తిగా అంగీకరించను
- 2 = అంగీకరించను
- 3 = తటస్థం
- 4 = అంగీకరిస్తున్నాను
- 5 = పూర్తిగా అంగీకరిస్తున్నాను

29. పశువుల మార్కెట్లలో సరైన (నిజమైన) కొనుగోలుదారులు లేదా అమ్మకందారులను కనుగొనడం కష్టం.

30. పశువుల వ్యాపారంలో సరైన ధర పొందడానికి బేరసారాలు అత్యంత అవసరం.

31. కొనుగోలు చేయడానికి ముందు పశువుల నాణ్యతను అంచనా వేయడం కష్టం.

32. ప్రస్తుత పశువుల వ్యాపార విధానంతో నేను సంతృప్తిగా ఉన్నాను.

33. స్మార్ట్ఫోన్ అప్లికేషన్లను ఉపయోగించడంలో నాకు పూర్తి నమ్మకం ఉంది.

34. డిజిటల్ పశువుల మార్కెట్ ప్లేస్ అనేది పశువుల వ్యాపారాన్ని మెరుగుపరుస్తుంది.

35. ధృవీకరించబడిన (వెరిఫైడ్) యూజర్ ప్రొఫైల్స్ పశువుల వ్యాపార యాప్ పై నా నమ్మకాన్ని పెంచుతాయి.

36. పశువుల వివరణాత్మక సమాచారం (జాతి, వయస్సు, బరువు మరియు ఆరోగ్యం) నేను మెరుగైన కొనుగోలు లేదా అమ్మకపు నిర్ణయాలు తీసుకోవడానికి సహాయపడుతుంది.

Appendix C: Oral Consent Script

This appendix reproduces, in English, the oral consent script that the researcher used, in Telugu translation, immediately before each field-study interview and questionnaire session in December 2025 and each usability-evaluation session in May 2026, addressing the gap identified in Section 2.6 and Section 6.2. The script was read aloud to each participant in Telangana Telugu before proceeding, and the researcher noted, for each session, that the script had been read and that the participant had verbally agreed to take part. No signed or written consent form was used, for the reasons discussed in Section 2.6; this script is offered as a record of what participants were actually told they were consenting to, rather than as a substitute written form.

C.1 Script Read to Field-Study Participants (December 2025)

- Purpose: “I am a student from PDPM IIITDM Jabalpur, studying how buying and selling of livestock happens at this Santha, so that I can design a mobile application that fits the way you already trade.”
- Procedure: “I would like to ask you some questions about your experience buying or selling livestock, and about your use of mobile phones. This will take about 15 to 20 minutes. There are no right or wrong answers — I only want to understand your experience.”
- Voluntariness: “Taking part is completely up to you. You do not have to answer any question you do not want to answer, and you can stop at any time, for any reason, without any problem to you.”
- Right to withdraw: “If you want to stop in the middle, or you want your answers removed after we finish talking, please tell me and I will not use them.”
- Confidentiality and data use: “I will not write down your name. Your answers will be combined with other participants' answers and used only for my thesis research at the institute. They will not be shared with anyone else, including other buyers, sellers or brokers at this market.”
- Risks and benefits: “There is no payment for taking part, and there is no risk to you beyond spending a few minutes of your time. Your answers will help design a better application for buyers and sellers like you in the future.”
- Confirmation: “Do you understand what I have explained, and are you willing to answer some questions?” (Participation proceeded only after a clear verbal 'yes'.)

C.2 Script Read to Usability-Evaluation Participants (May 2026)

The script used in the May 2026 usability-evaluation sessions repeated the purpose, voluntariness, right-to-withdraw and confidentiality elements above, adapted to the prototype-testing context, with the following additions:

- Purpose: “I have built a sample mobile application called Sukraven, based on what buyers and sellers like you told us last year. I would like you to try using it and answer a few questions about your experience.”
- Procedure: “I will ask you to complete a few tasks on this phone, such as looking at an animal listing or trying to bargain a price. Afterwards, I will read out some statements and ask whether you agree or disagree with each one. This will take about 20 to 30 minutes in total.”
- Nature of the prototype: “This application is not yet available to the public and no real animals, money or deliveries are involved today — everything you see is a sample for research purposes only.”
- Identification: “I will refer to you only by a participant number and whether you are a buyer or a seller when I write about this research; I will not use your name.”

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